POLITECNICO
MILANO 1863

COMPUTER SCIENCE AND ENGINEERING

SOFTWARE ENGINEERING II

2024 - 2025

RASD

Requirement Analysis and Specification Document

Students & Companies

Contents

1. Introduction

1.1 Purpose

The rapid expansion of the job market and the rising demand for skilled, reliable professionals with advanced education have made internships and apprenticeships valuable opportunities for university students to gain exposure to the professional world. Students may often feel overwhelmed by the laborious task of sorting through numerous internship opportunities to find those that best align with their study paths. On the other hand, companies offering a wide variety of internship programs find it challenging to select capable and hardworking students and to offer them opportunities that match their skill sets and fields of study. The purpose of S&C is to provide an environment where students and companies can connect, form professional relationships, and organize and manage internship programs in all their most relevant aspects. Users should have a comprehensive understanding of the entire matching process, from searching for opportunities to completing the internship, and transparency between students and companies should be fostered in every step of this journey.

1.1.1 Goals

G1: University students are assisted in finding projects and internship terms offered by companies that best match their skills, experiences, and interests, and in presenting themselves effectively to the companies they wish to engage with.
G2: Companies publish the projects and internship terms they offer and receive assistance in promoting these opportunities based on their professional fields. They are also supported in connecting with university students who might be suitable candidates for these roles.
G3: Companies communicate with students and subject them to selection processes for accessing the projects and terms they offer.
G4: Students and companies can provide feedback on the matchmaking process and the internship and communicate issues while the projects are ongoing, allowing universities to address them when needed.

1.2 Scope

The core function of the product is to act as a mediator in the various interactions between students and companies, and it also provides monitoring mechanisms not only for these two parties but also for the universities that the students attend. Thus, the external users who interact with the platform include companies, university students, and universities. Since companies and universities are registered organizations, the system must verify their reliability by introducing a fourth user, the S&C administrators[1], who are responsible for this procedure. All external users interact with the system through their accounts, which are personal for students or represent the entire organization for companies and universities. Students need to provide their CV to the system, which serves as a fundamental source of information for companies to assess their suitability for internship projects and for the system to enable the recommendation mechanism. Similarly, companies need to provide information regarding the internship projects they are interested in offering. The system facilitates communication between the two parties and supports the continuous decision-making process throughout all phases of matchmaking. If a company drafts a contract and proposes it to a student, and the student accepts, this sets the beginning of the collaboration. Users can monitor the internship throughout its duration, communicate with each other and with universities, and decide whether to end the ongoing term prematurely. Universities can also monitor their students' internships and address any complaints that arise.

1.2.1 World Phenomena

WP1: Students create a CV with information about their experiences, skills and attitudes.
WP2: Companies set up projects for students to participate in.
WP3: Students and companies conduct agreed-upon job interviews
WP4: Companies examine students' skills and interests based on their CV and decide whether to offer them a project opportunity
WP5: Students choose whether to apply for the projects proposed by companies.
WP6: Companies draft a contract for selected students.
WP7: Students decide to accept or decline contracts offered to them by companies.
WP8: During the course of a project, students and companies draw conclusions about each other.
WP9: Universities examine students' complaints and offer support during the internship period.
WP10: S&C administrators verify whether an organization requesting to register is reliable.

1.2.2 Shared Phenomena

SP1: Students publish their CV.
SP2: Companies publish available projects.
SP3: Students look for available projects.
SP4: Companies look for available students.
SP5: The system provides suggestions to companies and students on how to make appealing submissions.
SP6: Students choose to follow or ignore recommendations about suitable companies provided by the S&C platform.
SP7: Companies choose to follow or ignore recommendations about suitable students provided by the S&C platform.
SP8: Students and Companies communicate during the selection process.
SP9: The system informs students about possible internships they could be interested in.

SP10: The system informs companies about students that are suitable for a project.
SP11: Students and companies provide feedback to each other, and send complaints.
SP12: The system collects feedback about the selection process provided by students and companies and uses it for recommendation analyses.
SP13: The system collects complaints provided by students and companies during the internship period and provides them to the respective university.
SP14: S&C administrators are provided by the system with the information of organizations that are requesting to register.

1.3 Definitions, Acronyms, Abbreviations

1.3.1 Definitions

- **Internship**: Temporary position held by a student for a specified period, allowing them to work under the supervision of professionals in a company. Internships can be paid or unpaid and offer both tangible and intangible benefits, such as training and mentorship.
- **Recommender system**: A mechanism that alerts students when a new project of interest becomes available and notifies companies when a new CV matching their requirements is published. It incorporates statistical data derived from feedback provided by students and companies during internships. This data is analyzed and utilized to enhance the system's effectiveness in making recommendations.
- **Matchmaking process**: The process that begins when a student and a company connect and concludes when the final contract is signed or one of the parties decides to discontinue. It includes communication through the platform, the interview process, and the proposal phase.
- **Handshake**: The first connection request carried out by one of the two parties. If accepted it sets the beginning of the matchmaking process.
- **Explore:** The section of the platform that offers support to both students and companies in finding valuable opportunities, and that provides recommendations and notifications about positions and CVs uploaded on the system.
- **Legal Documentation:** the collection of official documents that describes an organization’s legal status and ensures its reliability.

1.3.2 Acronyms

- **S&C**: Students and Companies.
- **CV**: Curriculum Vitae.

1.3.3 Abbreviations

- **G**: goal

- **WP** : World Phenomena
- **SP** : Shared Phenomena
- **IdP**: Identity Provider

1.4 Revision History

- Version 1.0: 17/12/2024

1.5 Reference Documents

The assignment for this document and all the information included refer to the following documentation:

- The specification for the 2024/25 Requirement Engineering and Design Project for the Software Engineering II course.
- The slides on the webeep page of the Software Engineering II course.

1.6 Document Structure

1. **Introduction**: a brief description of the project that follows from the analysis of the specification provided, it highlights the goals of the product and its scope.
2. **Overall Description**: a more detailed description of the product that comprehends also real world scenarios and analyses the system from a technical point of view
3. **Specific Requirements**: a detailed list of all the requirements needed in order to satisfy the goals, and their relationship with domain assumptions
4. **Formal analysis using Alloy**: a formal description of a portion of real world phenomena analyzed through Alloy in order to highlight inconsistencies between assumptions and phenomena previously modelized.
5. **Effort spent**: the time spent by each member of the group to contribute to the realization of this document.
6. **References**: reference to documents or tools used for writing this document.

2. Overall Description

2.1 Product perspective

2.1.1 Scenarios

Student edits his profile information

Student S wants to find suitable jobs based on their profile. They enter the S&C platform with their university mail, and once logged into the platform, they select “My profile” in the menu. They press the icon to upload the CV if it hasn't been uploaded yet, or replace the current CV with a new one if they wish to update it. The platform also shows tips to make their current CV more appealing, so they can follow those suggestions when creating a new version. They can edit their information, such as locations, by pressing the pencil icon. They can also toggle their CV to public (if it is not already active), allowing their profile to be included in company searches and recommendations.

[In this example, we considered a Student; as explained in the “Use Cases” paragraph (3.2.2), Companies can also modify their profile.]

Company publishes a new project

Company C organized a new project P and wants to publish it on the platform to hire students suitable for it. After logging in, C goes into its profile section and clicks the button to edit its projects C presses the button to add the project, and adds the information about P: its title, its location, a descriptive text, and one or more files. Afterwards, C saves the modifications and project P becomes visible to all students. The platform will also identify suitable students for the project and notify them accordingly.

Student looks for a new internship

Student S wants to find a new internship to acquire more skills and experience. They log into the platform and then go into the “Explore” section. Here they will already find the internships that the system detected as suitable for them based on their CV. They browse the recommendations and click on those that catch their attention. If the recommendations do not meet their expectations, they can switch to a global search, adjust filters, and continue exploring. Clicking on an internship reveals detailed information about the associated project P, which is posted by company C. If interested in the explored offer , student S sends a handshake to C. Then, C can accept the handshake and now, project P involving S and C will now appear in the “My Internship” tab to both company C and student S.

[In this example, we considered a Student; as explained in the “Use Cases” paragraph (3.2.2), Companies can also search and browse through the available and recommended students.]

Student sends a message to a Company

Student S is going through the selection process for project P with company C and has questions regarding some ambiguous details in the project description. Therefore, the best way to address these concerns is by sending a message to C and asking about their doubts. S goes to the “Messages” section, and, since the selection process with C has already started, they will already find here the chat with C containing the messages history. They select this chat and type a message. While they’re waiting, they don't have to stay on the platform, as they will receive a notification once C replies.

[In this example, we considered a Student; as explained in the “Use Cases” paragraph (3.2.2), Companies can also send messages.]

Selection Process

- **Company sends a handshake**

Company C opens the Explore tab and, among the recommendations, finds a student, S, that captures its interest for project P. C reads their CV and their information and decides to reach out. Therefore, C sends a handshake message to S, specifying the project for which they are interested in S and including a greeting. Student S receives a notification saying that a handshake message has arrived. They go into the platform, read the message, and respond positively, indicating their interest in proceeding with the selection process. The internship regarding project P involving S and C will now appear in the “My Internship” tab to both company C and student S.

[In this example, we considered a Company; as explained in the “Use Cases” paragraph (3.2.2), Students can also send handshakes.]

- **Company interviews a Student**

Once company C and student S have completed the handshake phase, C has to decide whether to admit S in its project or not. C prepares a questionnaire and a video call for the evaluation. It can post the related links in the corresponding internship page that now appears in the “My Internship” section of the platform. C can contact the student through messages to decide a suitable day and time. The student will click on the links and answer the questionnaire, and then will join the video call at the established time. After an internal meeting to evaluate S’s suitability, it communicates through the platform whether C is interested in hiring him, and S receives a notification about it.

- **Company makes the Student a contract proposal**

The platform will now show to the company C an upload button in the internship page where the contract proposal can be uploaded. The company prepares it and uses that button to upload it. Student S will receive a notification and will view the contract. After noticing a scheduling conflict - Friday workdays overlap with an academic project - S contacts C via message to discuss the issue. Company C acknowledges it and, in the same internship page, clicks the button to upload a modified version of the contract. S receives a notification, reviews the updated terms, and formally accepts the offer through the platform. The internship is now officially underway.

Student sends a complaint

Student S is attending an internship with company C. The contract outlined that S would be directly involved in software development, supervised by experienced coworkers to acquire new skills. However, the company has not assigned S any relevant task yet. After unsuccessful discussions with their manager, S decides to send a complaint to his university. They log into the S&C platform, select the corresponding internship in the “My Internships” tab, and here they will find a section that allows them to send a complaint to their university, U. They write the message and click “send”. U reads the complaint and decides to talk directly to the manager of C through a phone call.

[In this example, we considered a Student; as explained in the “Use Cases” paragraph (3.2.2), Companies can also send complaints.]

Company terminates a contract of a student

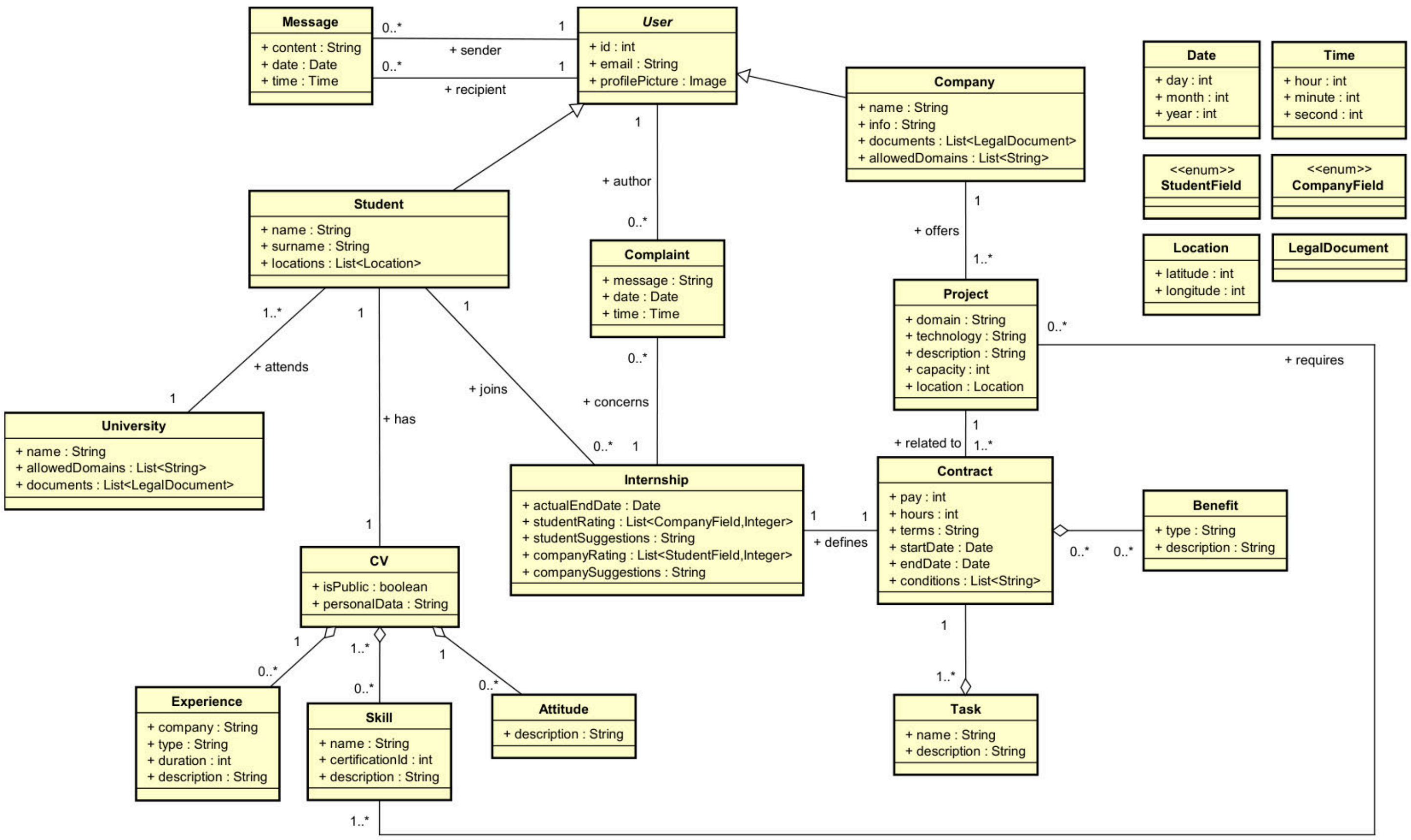
Company C is not satisfied with the work of student S. They always come to work late, despite prior warnings and a formal complaint to S’s university. Since C can’t afford to lose more time, it decides to terminate the contract. They log into the platform, select the corresponding internship in the “My Internship” section, and use the area dedicated to contract termination. C writes its explanation, clicks the send button, then the system asks for confirmation; after doing so, the message is sent to both the student S and their university. Now that the internship is over, a satisfaction survey pre-computed by the platform will become available to both C and S.

[In this example, we considered a Company; as explained in the “Use Cases” paragraph (3.2.2), Students can also decide to terminate their contract.]

Student and Company fill out the satisfaction survey

Student S has completed their internship with company C. The project was engaging, and they acquired new skills. Additionally, C is satisfied with S's performance at the workplace. Once the internship ends, the platform notifies C and S that the satisfaction survey is available. In the “My Internships” tab, they select the corresponding internship, and here they will find the survey. After clicking the “open” button, they will both have to rate each other based on standard fields with stars from 1 to 5, and they can also leave a textual feedback: for example, S wants to suggest C to adopt a more structured way of assigning tasks to their employees, while C suggests S to be more proactive. Once they’re done filling the survey, they press the button “send”. They will both receive a notification and will be able to see each other’s answers. The internship is now no longer visible in the “My Internship” section.

2.1.2 Domain Class Diagrams



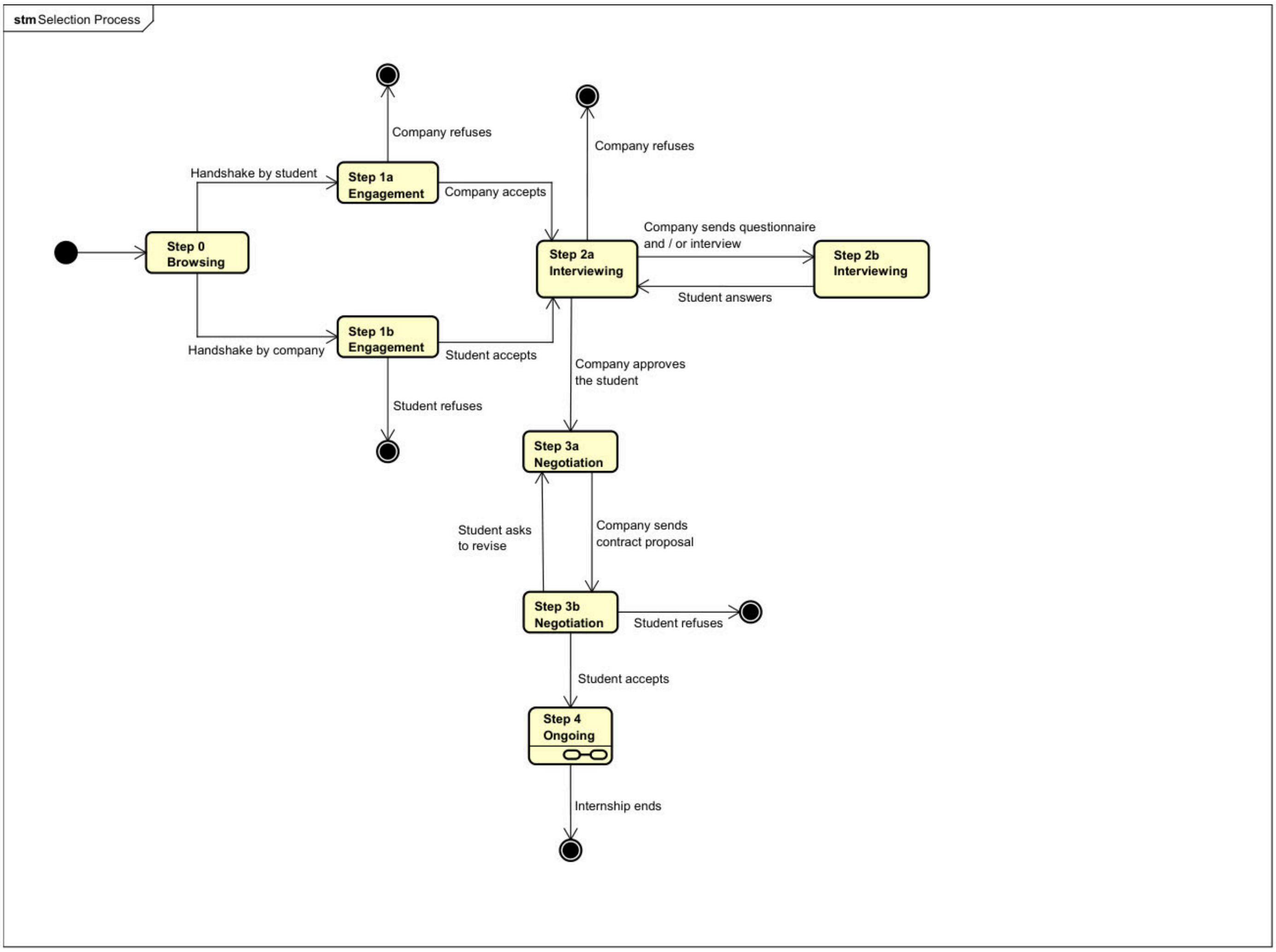
This Domain Class Diagram represents the entities involved in the domain of our system and the relationships between them. Some important points to note:

- In this diagram, we do not go into the details of the selection process; we only represent internships that are already ongoing or have been completed, with contracts already signed.
- The University is technically also a User of our system; however, since it is a "special" user and cannot be the author of complaints and messages, it is not included as a subclass of the "User" entity.
- StudentField and CompanyField represent the set of fields evaluated in the Satisfaction Survey completed at the end of the internship by both parties. A Student rates the company with an integer from 1 to 5 for each field in CompanyField and can leave a textual suggestion (studentSuggestion); vice versa for the Company. The values will remain 0 until the survey is completed.
- In the Company and in the University entities, the attribute “documents” contains a set of legal documents allowing S&C administrators to verify whether the organization is reliable or not.
- In Internship, actualEndDate might differ from the endDate in the Contract because one of the parties could decide to terminate the internship prematurely.

- Complaints are to be considered inside the context of a single internship and one between the Student and the Company involved in the internship is the author of it. In both cases, the complaint is forwarded to the university.
- Messages can only be sent from a Student to a Company and vice versa, not between users of the same group.

2.1.3 State Diagrams

Selection Process

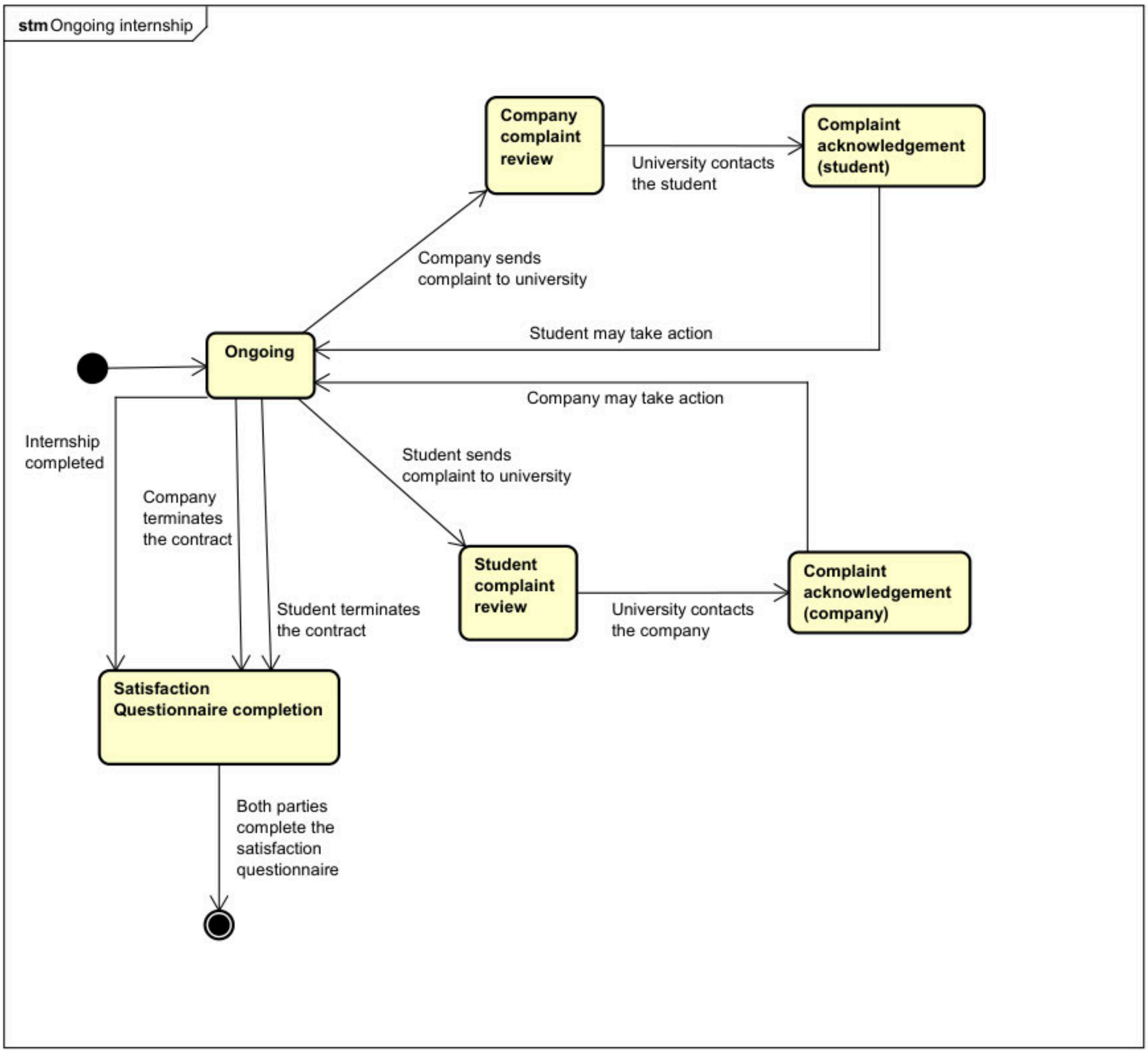


This state machine diagram describes the steps in the selection process. The step number will be displayed to users, while the letter is used to distinguish different states within the same conceptual step.

At the beginning, every Student-Company pair starts at Step 0. Both parties can send a handshake to each other, which moves them to Step 1. If the student initiates the handshake, their CV will become visible to the company, even if it had chosen to keep it private.

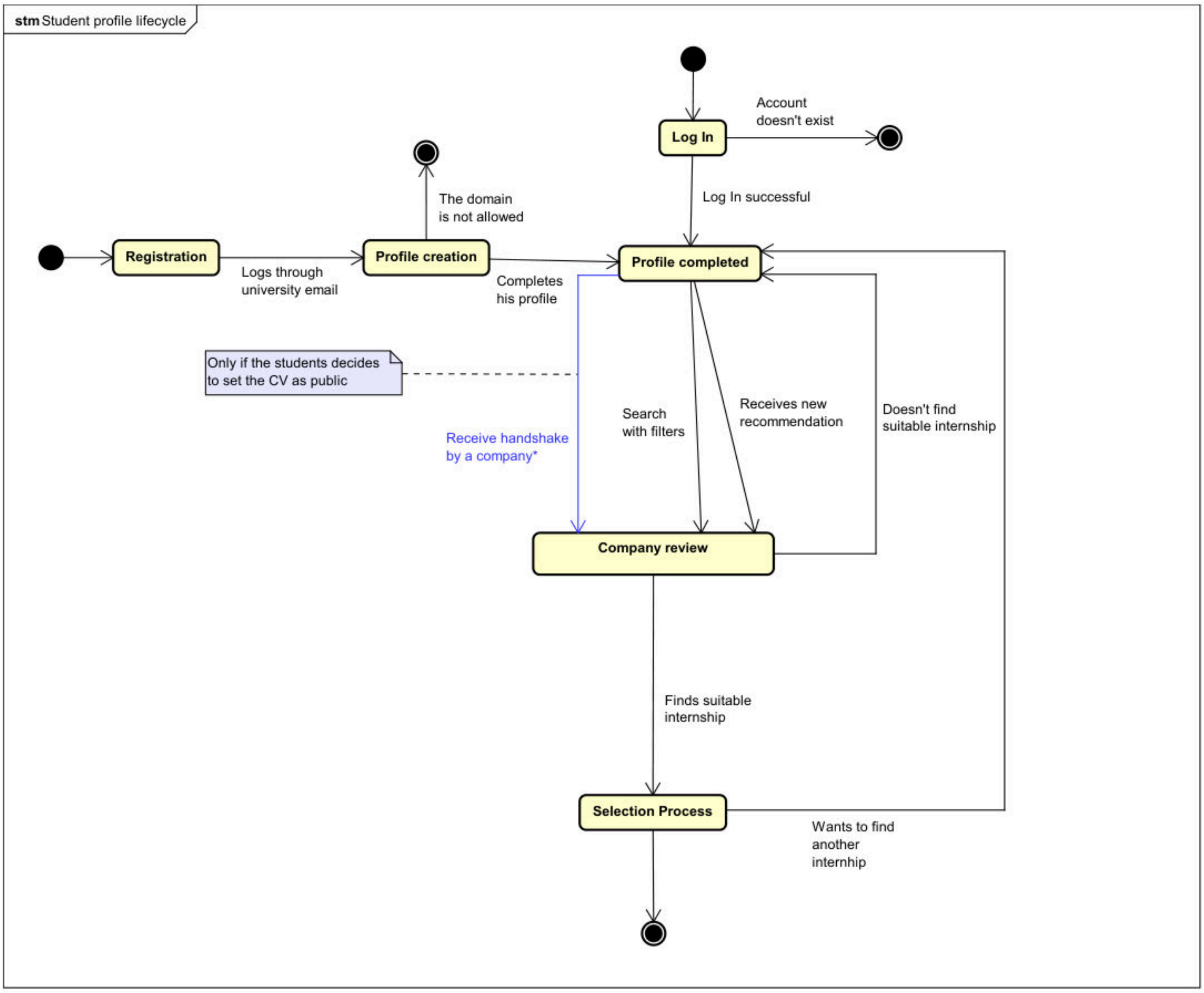
If the recipient of the handshake decides to accept, Step 2 begins. During this phase, the company can send questionnaires to the student to answer and schedule an interview. Once the company approves the student, they send a contract proposal (Step 3); the student can choose to accept, refuse, or request a revision for better conditions. If the student eventually signs the contract, the internship begins (Step 4).

Ongoing Internship



This state machine diagram describes the possible states of an ongoing internship, particularly regarding complaints. Both parties can submit complaints to the student's university, which is responsible for addressing them by communicating directly with the other party. This interaction will take place outside our system (e.g., via email or phone call). If either the student or the company decides to terminate the contract, the university and the other party will be notified and provided with a description of the reason for the decision.

Student Profile Lifecycle

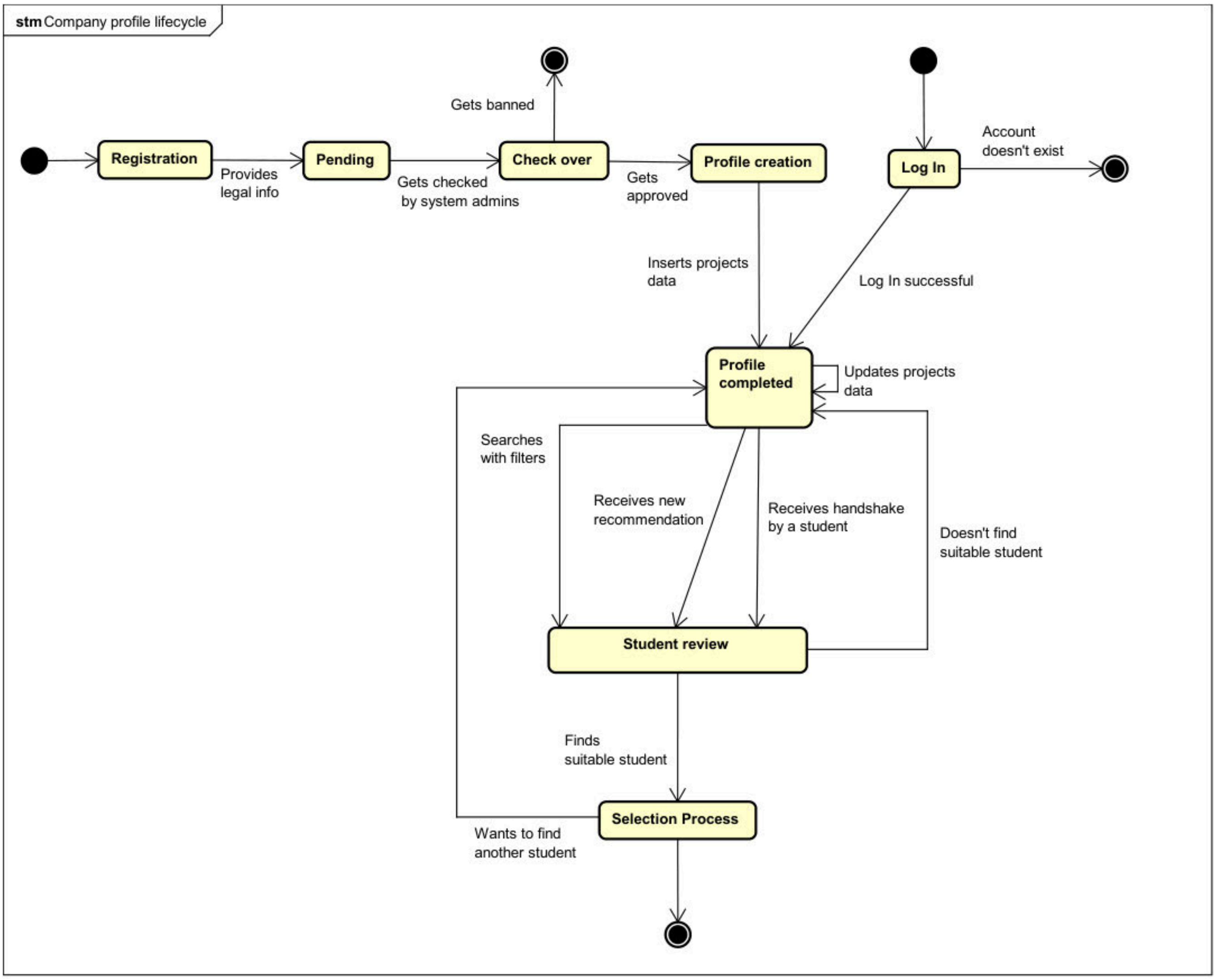


This state machine diagram represents the actions a student can take with their profile outside the context of an internship (described in other diagrams). The subject of every transition is always the student. A student can choose to keep their CV private: in this case, they can send handshakes to companies, and only those companies will be able to see the CV. Other companies will not be able to find the student through search or recommendations.

The profile can be made public at any time, making the student and their CV visible to all companies and potentially recommended by the system. Therefore only in this case the student could receive handshakes from companies (blue arrow).

There are two initial states that correspond to Registration (only the first access) and Log In [further explanation on how the authentication works in the "Use Cases" paragraph (3.2.2)].

Company Profile Lifecycle



This state machine diagram represents the actions a company can take with its profile outside the context of an internship [already described in the other diagrams]. The subject of every transition is always the company. To create a profile on S&C, a company must provide legal documentation that will be verified by S&C system administrators to ensure its reliability. The company profile is always public.

There are two initial states that correspond to Registration (only the first access) and Log In [further explanation on how the authentication works in the "Use Cases" paragraph (3.2.2)].

2.2 Product Functions

Sign up & Login

The registration and login functionalities for organizations differ in several ways from those for students; therefore it's better to separate them based on the intended user group.

- Students: To sign up for the system, a student must provide their name and an email address associated with their university domain. The system verifies that the email matches a registered domain and then redirects the student to the university's authentication system. Once authenticated, the student is brought back to the platform and given a form to enter their preferred job position and location. Finally, the profile is created, and the student's information is saved in the system database.

  To log in, a student presses on the login button, enters their email and gets redirected to the university access page. After authenticating with their university credential, returns to the system home page and is correctly logged in.

- Organizations: To register to the platform, an organization must provide legal information to verify its reliability. Depending on the organization, the system requires to insert its legal documentation and a valid email domain, then redirects the user to a page to await the completion of verification procedures. The verification procedure is delegated to the administrators, who are responsible for checking the reliability of the user and approving the registration request.

  To log in, an organization operator enters an email address associated with the organization domain and is redirected to the organization access page. After authenticating with their credentials and receiving permission to access the organization account, the operator is taken to the home page and is successfully logged in.

Profile enhancement (for Students)

Students can enhance their profiles by adding detailed information such as skills and academic achievements. The platform provides recommendations on what's missing, such as specific skills, experience, or known technologies, to make their profile more appealing to companies. Students can upload their CVs and list their skills, receiving suggestions to complete their profiles for better visibility and job matching.

Job Postings Improvement (for Companies)

Companies can create more attractive job postings by providing detailed descriptions, clear requirements, and company culture insights. The platform suggests keywords for job titles and descriptions to increase discoverability and recommends missing details, such as location preferences or salary ranges, ensuring the job posting is as complete and appealing as possible. Additionally, companies can target specific student profiles based on their skills and interests, ensuring better job matches.

Search of suiting candidates

Both students and companies can find suitable profiles on the explore page. Students can search for Job listings that match their skills, interests, and career goals, while companies can discover candidates whose profiles align with job requirements, skills, and qualifications for each provided project. The explore page allows users to perform a global search and filter results based on relevant criteria, helping them easily find the best matches for their needs; it also shows personalized recommendations based on their profile, detected by the system. Both users can send requests to the selected profile and create a connection with them.

Selection Process & Internship Management

- **Recruitment Management (for Companies)**

Companies can explore all the candidates they have connected with, or who have sent requests, grouped by project. They can conduct questionnaires with selected candidates to assess their suitability. Afterward, companies can negotiate contracts with students. At each step of the process, companies have the option to either reject a candidate or proceed with the recruitment, ensuring a streamlined and efficient hiring flow.Finally they can set the internship as ongoing.

- **Job Offers Management (for Students)**

Students can browse the jobs in their personal page, no matter if they were contacted by the company or if they applied for the job themselves. They choose to participate in all the interviews or questionnaires they were invited to. Afterward, they can negotiate the contract conditions until they are satisfied with the terms and choose to accept the offer.

- **Ongoing Job Management**

Both students and companies can actively participate in managing the ongoing job or internship. Students can track their progress, communicate with the company, and receive feedback, while companies can monitor performance, provide guidance, and assess the student's development throughout the duration of the job.

Satisfaction Survey

At the end of an internship, the system asks both parties to fill up a satisfaction survey. Students and Companies will be asked to rate each other based on some predefined fields using a scale from 1 to 5. The survey also contains a space to provide feedback and suggestions. The new ratings will contribute to the average rating displayed on profile pages.

University Oversight

Universities can monitor the selection processes and the ongoing internships of their students. Moreover, students and companies have the option of sending a formal complaint to the students' university. Once the universities receive the communication, they can decide independently how to behave. Any further interaction will take place outside the context of the system.

2.3 User characteristics

2.3.1 Student

The Student is able to register and login to the platform. His profile contains his personal details, such as location, university and CV.
He can make his profile visible for companies while he wants to be contacted with job proposals, or maintain it private if does not intend to share his details. He can search for jobs by tuning search filters as he wishes, or can browse the jobs that were recommended to him. He is able to apply for jobs, and keep in touch with the company while the selection process is ongoing, while negotiating job conditions, and afterwards in case the job is started. He has to complete a satisfaction survey in the case he finishes the internship or if he interrupts it before its official end.

2.3.2 Company

The Company is able to register and login to the platform. Its profile contains a description, location, and info about the projects. The company can create projects and edit them. They automatically become available for students registered on the platform until the company chooses to delete it. The company receives recommended students for each job available and is able to visualize those student profiles. Also the company is able to search students globally. It can send invitations to students , and can maintain communication throughout the entire selection process and while the internship is ongoing. It has to complete a satisfaction survey when the internship is over.

2.3.3 University

The University does not have a public profile, as it is not directly involved in internships, but it still has an account. It can browse through its students' internships to monitor them, and is responsible for addressing complaints submitted by companies or students in the context of internships where one of its students is involved.

2.3.4 S&C Administrator

S&C Administrators are responsible for verifying the reliability of companies and universities by analyzing the legal documents submitted during the registration phase. They have the authority to either approve or reject these users. Only students from approved universities will be accepted.

2.4 Assumptions, dependencies and constraints

2.4.1 Regulatory policies

The S&C platform requests students to provide their university email, full name, date of birth, preferred locations, and, most importantly, their CV. It is essential that this data, along with private information about internships and exchanged messages, are handled with absolute privacy and in compliance with the General Data Protection Regulation (GDPR). Additionally, this information must not be used for commercial purposes. Furthermore, companies and universities are required to submit legal documents to verify their credibility. These documents must be treated with due confidentiality.

2.4.2 Domain Assumptions

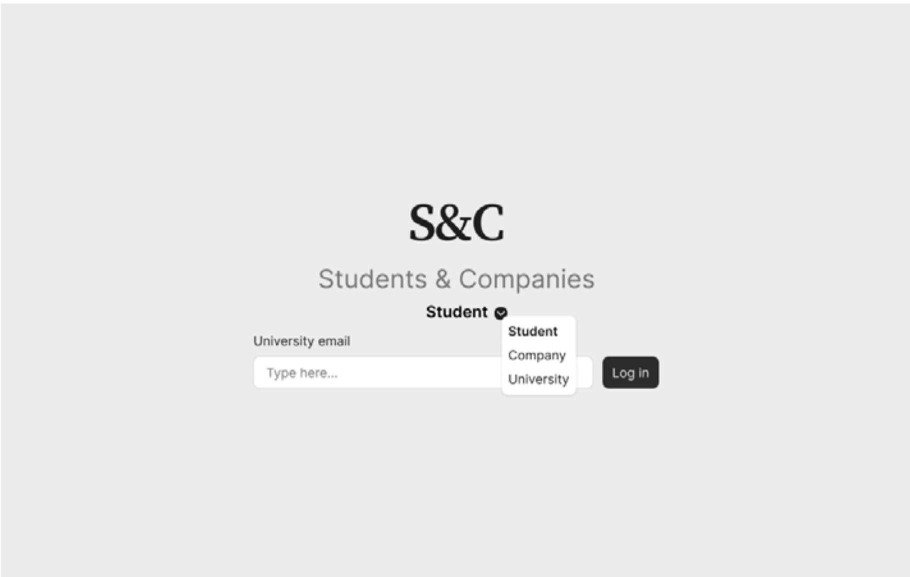
- D1: Students' personal information is accurate.
- D2: S&C administrators are able to verify through external checks whether companies and universities requesting to register are reliable.
- D3: Students write their CVs, which contain correct information and include the description of their experiences, skills and attitudes.
- D4: Companies provide detailed information about the projects they offer and related contract terms for students.
- D5: Students and companies check the recommendations provided by the system and use the platform to accept suitable matches.
- D6: Companies prepare structured questionnaires and interviews to which they subject students.
- D7: Students and companies are open to communicating and providing feedback on each other.
- D8: Students are enrolled at the university they specify when registering to the platform and are provided with a personal university email.
- D9: Universities monitor internships and handle complaints that may require their interruption.
- D10: All users have a reliable internet connection.

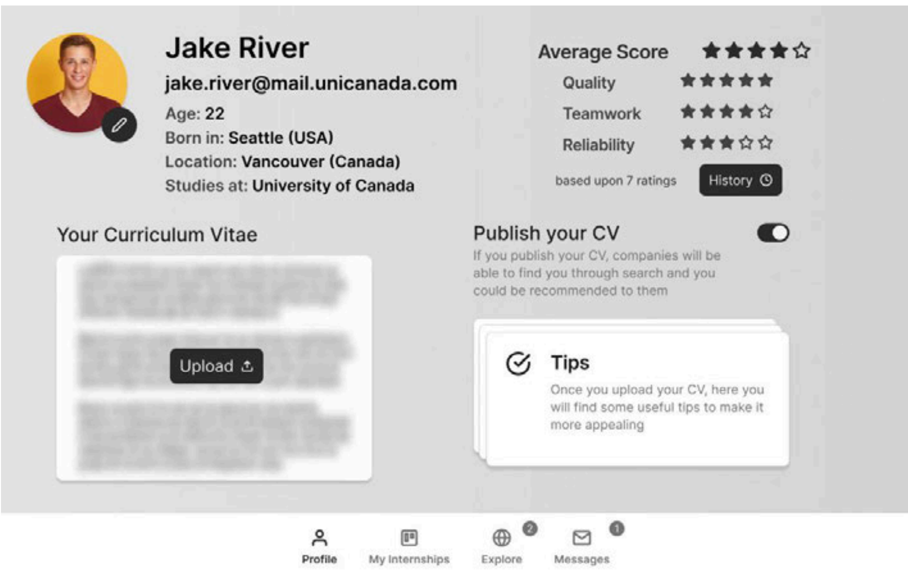
3. Specific Requirements

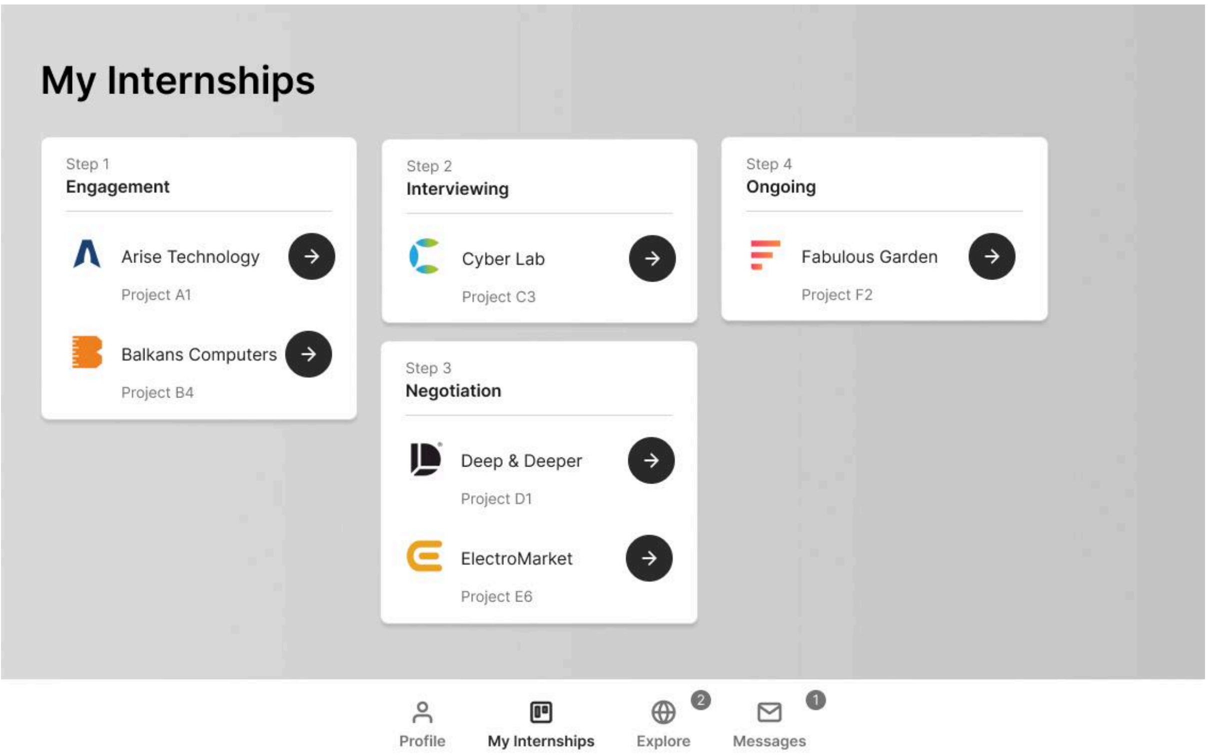
3.1 External Interface Requirements

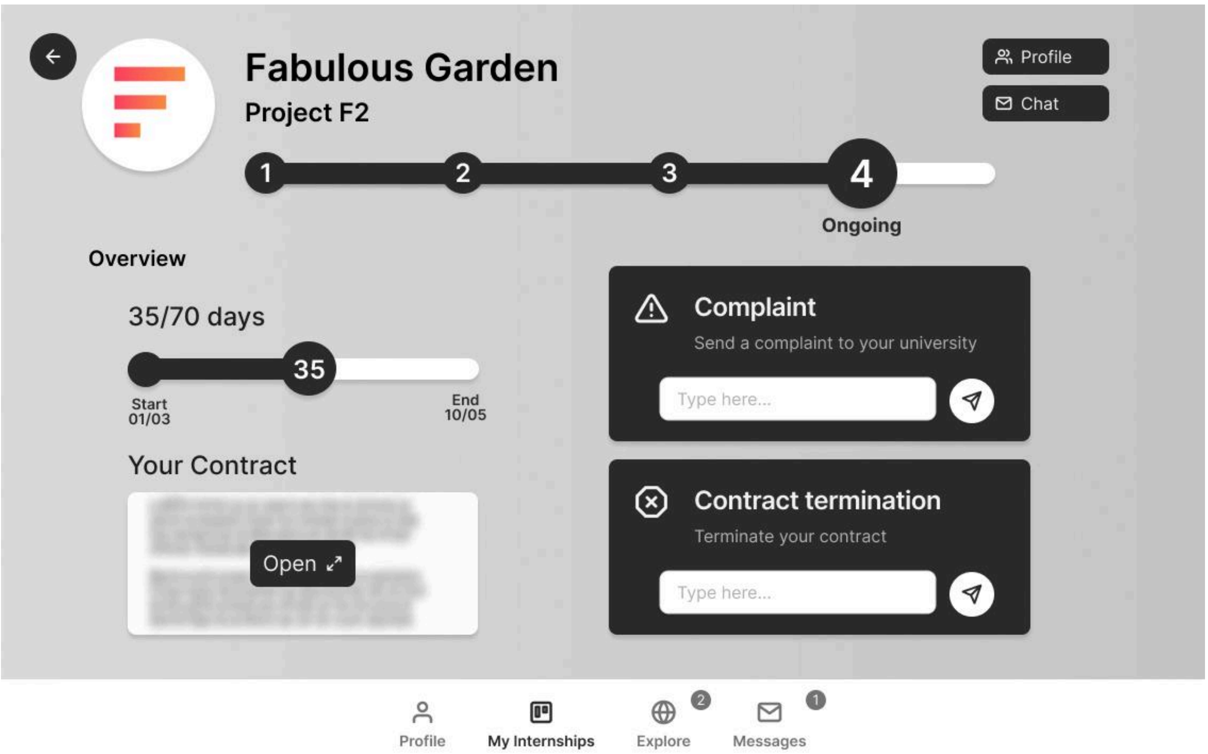
3.1.1 User Interfaces

Here is shown a mockup of the web application UI. The images outline the user interface of the most important screens from the point of view of the student. The point of view of the company is quite similar; the main differences are the profile section (in which, instead of the CV, companies upload the information and documents about the projects) and the type of user the system shows them in the explore section and in chats. Universities have a smaller user interface, limited to monitoring the internships of their students (with a view similar to "My Internships" section).






My Internships
Step 1
Engagement
Arise Technology
Project A1
Balkans Computers
Project B4
Step 2
Interviewing
Cyber Lab
Project C3
Step 3
Negotiation
Deep & Deeper
Project D1
ElectroMarket
Project E6
Step 4
Ongoing
Fabulous Garden
Project F2
Profile
My Internships
Explore
2
Messages
1


Fabulous Garden
Project F2
Profile
Chat
1
2
3
4
Ongoing
Overview
35/70 days
35
Start
01/03
End
10/05
Your Contract
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Terminate your contract
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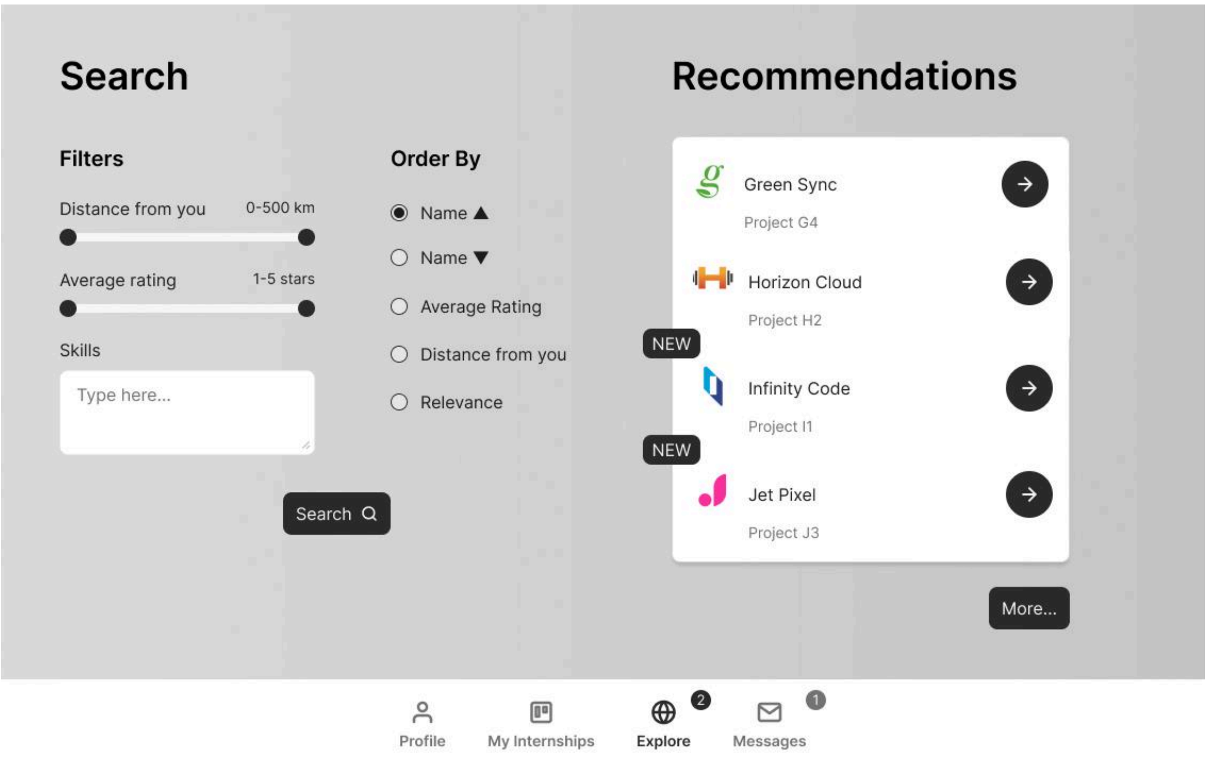
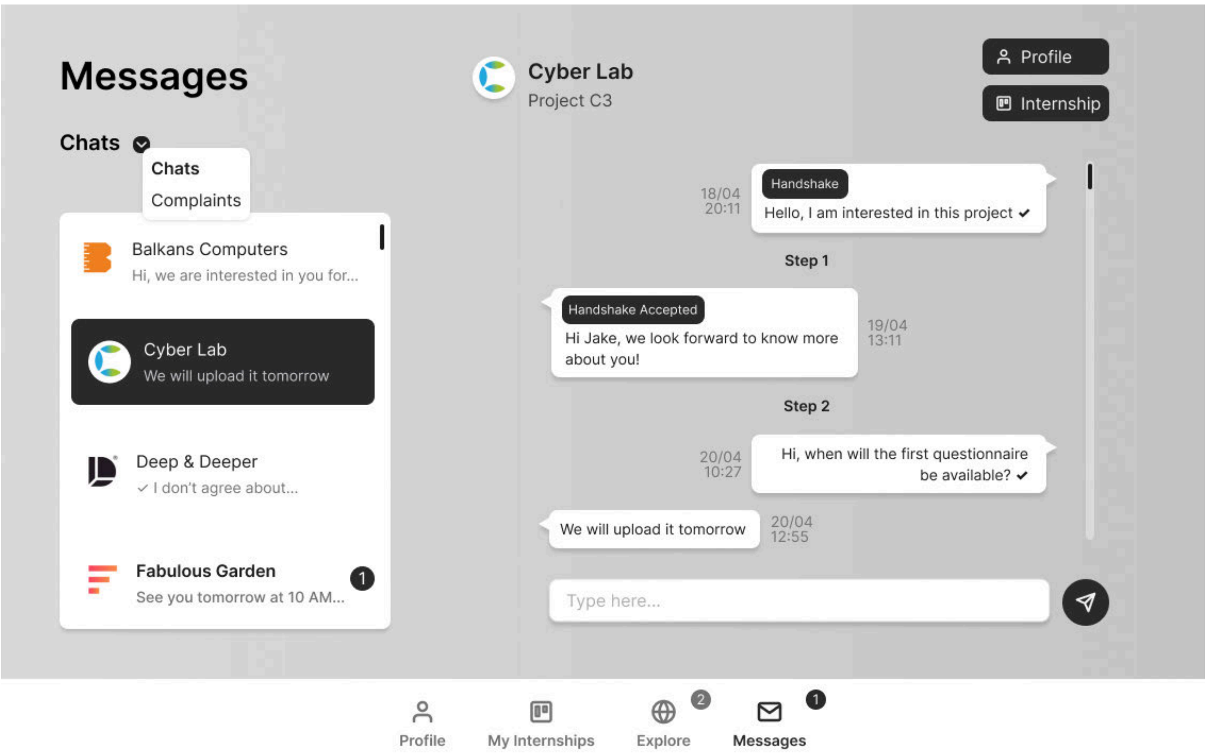
My Internships
Explore
2
Messages
1





3.1.2 Hardware Interfaces

Our platform is a web app, requiring only a device with a web browser, such as a computer, smartphone, or tablet, and an internet connection. No additional hardware is necessary, though optional devices like cameras or microphones may enhance certain features.

3.1.3 Software Interfaces

- **Identity Provider Interface:** to enable users to access through a university or a company account, the system relies on an identity provider. The identity provider is responsible for linking the organization account to the S&C account and allows the platform to retrieve identity information when a registered user logs in.
- **CV analysis tool:** The system must be able to utilize information from any submitted CV, thus it requires a tool to analyze uploaded CVs and extract all relevant data that refers to students' skills and interests. This data can then be used to better position the student within the platform's search mechanism.

3.1.4 Communication Interfaces

The S&C platform, being a web application, must communicate with users over the web. The interactions, including data transmission and document uploading, must be secure and encrypted using HTTPS. Authentication is managed using Single Sign-On (SSO). The platform must also use an appropriate protocol for managing the chat feature (e.g., WebSocket). Additionally, the application should support automatic email sending for notifications.

3.2 Functional Requirements

Sign up & Login

- R1: The platform allows university students to create an account.
- R2 : The platform allows companies to create an account.
- R3 : The platform allows universities to create an account.
- R4: The platform is able to verify the reliability of its registered users.
- R5: The platform allows registered users to log in.

Profile enhancement (for Students)

- R6: The platform allows users to update and modify information in their profile.
- R7: The platform allows students to upload and publish their CVs.
- R8: The platform offers suggestions to students to enhance the appeal of their CV.

Job Postings Improvement (for Companies)

- R9: The platform allows companies to publish available project offers along with all related details and information.
- R10: The platform offers suggestions to companies to refine the presentation of their projects.

Search of suiting candidates

- R11: The platform allows students to proactively look for internships and provides filters to help them refine their searches.
- R12: The platform analyses the submissions and computes recommendations that presents to both parties.
- R13: The system notifies students when a new project that could interest them gets published on the platform by a company.
- R14: The system notifies companies when a new CV that could interest them gets published on the platform by a student.
- R15: The platform allows students to visualize companies' profiles.
- R16: The platform allows companies to visualize students' profiles.
- R17: The platform allows students to search for job listings based on their skills, interests, and career goals.
- R18: The platform allows companies to discover potential candidates whose profiles align with job requirements, skills, and qualifications for each project.

Selection Process & Internship Management

- R19: The platform allows companies to submit questionnaires to students they are interested in and organize interviews with them.
- R20: The platform allows companies to offer contracts to students they wish to engage in one of their projects.
- R21: The platform enables direct communication between students and companies through a messaging system.
- R22: The platform allows companies and students to withdraw from the selection process and internship phase at any time.
- R23: The platform provides students and companies with tools for tracking and monitoring the matchmaking processes and the internships they are involved in.

Satisfaction Survey

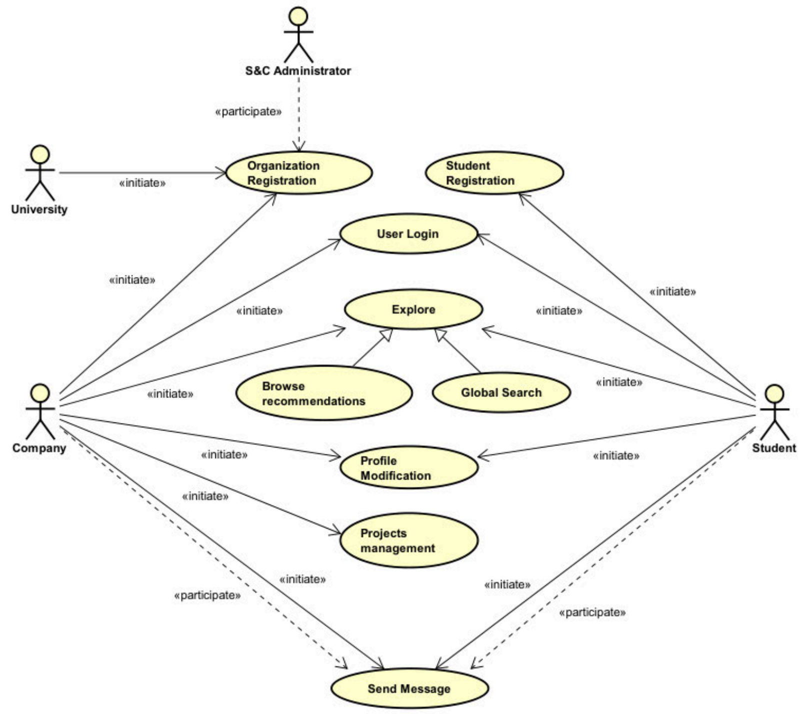
- R24: The platform collects feedback and suggestions from both students and companies and uses this information to feed the analysis mechanisms used for the recommendations.

University Oversight

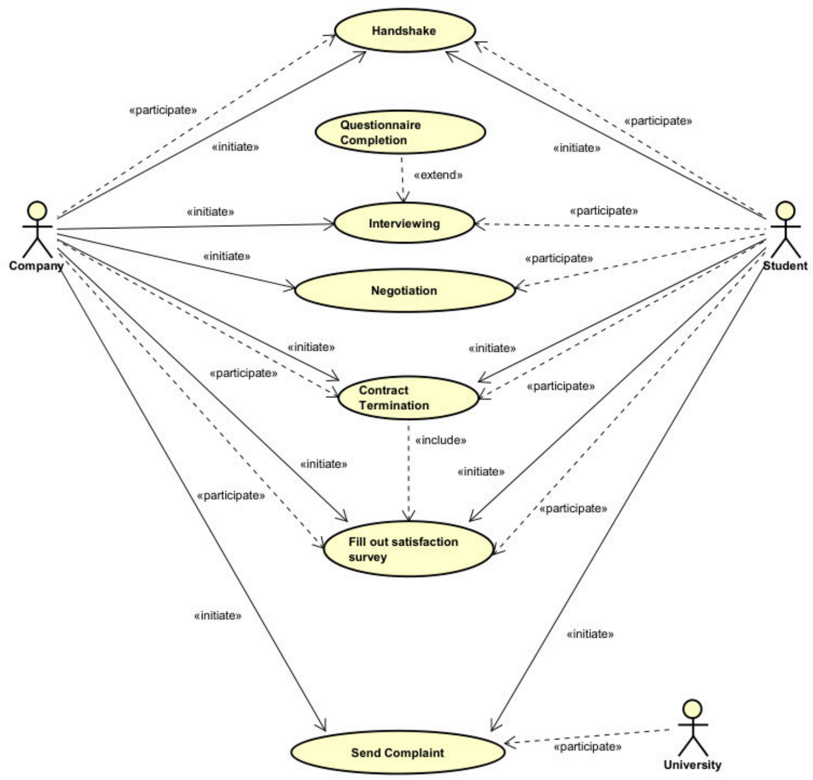
- R25: The platform allows universities to visualize companies' and students' profiles.
- R26: The platform equips universities with tools to monitor matchmaking processes and the internships in which their enrolled students participate.
- R27: The platform provides dedicated spaces where universities can view any complaints from both companies and related students about ongoing internships.

3.2.1 Use Case Diagram

ACCOUNT & UTILITIES



INTERNSHIP MANAGEMENT



3.2.2 Use Cases

[UC1] Organization Registration

Name	**Organization Registration**
Actors	Company operator - University operator, S&C administrators
Entry Condition	User opens the S&C platform
Event Flow	(a) The user presses "University Sign up" or “Company Sign up”. (b) The system requires the user to fill out a form with the organization's name and location. Then the user is required to provide the organization’s legal information and one of its valid domains. (c) The system submits the registration request to S&C administrators and requires the user to wait for the verifications procedures to be completed. (d) S&C administrators verify the legal information provided and complete the organization registration.
Exit Condition	The user has successfully registered to the platform
Exception	(d) S&C administrators are not able to verify the reliability of the organization

The only difference between company and university registration use cases are the set of legal information that the system requires to be submitted. The two functions are here considered as one for simplicity, but in the platform they are splitted, as shown by the two different sign up buttons.

[UC2] Student Registration

Name	**Student Registration**
Actors	Student
Entry Condition	User opens the S&C platform
Event Flow	(a) The user presses "Sign up" to begin. (b) The system displays a form to enter personal details: student’s full

	name and university email. (c) The system verifies that the university associated with the email is registered and that the inserted email is not already present, then forwards the student to the university authentication system. (d) The student authenticates with university credentials. (e) The student is redirected to the platform. (f) The system requires the student to enter desired job positions and preferred location (g)The information is saved and the profile is created. The student gets redirected to the login tab.
Exit Condition	The Student has successfully registered to the platform.
Exception	(c) The email inserted is not associated to any registered university (c) The email is already present in the system database (c) The student cannot be redirected (d) The student is not able to authenticate through the university system

[UC3] User Login

Name	**User Login**
Actors	Student, Company, University
Entry Condition	User opens the S&C platform
Event Flow	(a) The user presses "Login" to begin. (b) The user is required to enter their email. (c) The system redirects the user to their organization portal. Students and university operators are redirected to the university login page, while company operators are redirected to the company login page. (d) The user enters their credentials and gets redirected to the platform (e) The system shows the main page dashboard

Exit Condition	The User has successfully logged in to the platform
Exception	(b) The provided email is not associated with a registered account (c) The user cannot be redirected

[UC4] Explore

Name	**Explore**
Actors	Student, Company
Entry Condition	User is logged into the S&C platform
Event Flow	(a) The user opens the Explore page, and their recommendations are automatically displayed. (b) If the user wants to perform a global search, they type keywords in the search bar and press the search icon. (c) The system fetches results based on the keywords and displays relevant items. (d) The user can apply filters (e.g., location) to refine both the recommendations and the search results. (e)The user selects an item from the results to view its details. (f) The system displays the item's details, including additional actions (e.g., apply)
Exit Condition	The user successfully views the filtered results or details of a selected item

[UC5] Profile Visualization & Modification

Name	**Profile Visualization & Modification**
Actors	Student
Entry Condition	User is logged in the platform

Event Flow	(a) User presses Profile button (b) User presses “modify” button (c) To add preferred location: (i) The Student presses “add Location”. (ii) The System displays a search bar (iii) The student enters the preferred location (iv) The preferred location gets updated (d) To remove a location the student clicks on the ‘x’ next to the location to remove (e) To upload/replace the profile picture: (i) The student selects “upload profile picture”. (ii) The system allows the student to upload the picture (iii) The user uploads the profile picture (f) To upload / replace the cv (i) The student presses “Upload CV” (ii) The system allows the student to upload the CV. (iii) The CV is saved to the profile. Uploading a new CV causes the removal of the previous one. (iv) The system automatically extracts the student’s data from their CV and updates the recommendations. (g) Updated CV, location and profile picture are visible from the student’s page
Exit Condition	The user profile is successfully updated with the new details
Exception	(e) Picture size exceeds limit (f) File size exceeds limit

Name	**Profile Visualization & Modification**
Actors	Company
Entry Condition	User is logged in the platform
Event Flow	(a) User presses Profile button (b) The user presses the “modify” button. (c) The system switches the description field into an editable mode, allowing the user to edit the description text. Moreover, it is possible to update the profile picture. (d) The user makes the desired changes and presses the save button.
Exit Condition	The user profile is successfully updated with the new details
Exception	

[UC6] Project management

Name	**Project Management**
Actors	Company
Entry Condition	User is logged in the platform
Event Flow	(a) User selects the Profile Menu (b) Presses button “Manage Projects” (c) The system displays the list of existing projects with related descriptions and files previously provided (d) User can decide whether to update the data about one of those projects or to add a new project, providing the required information, such as textual description, locations, and files (e) The system displays the modified or created project in the users profile section and it becomes available for students in the explore page (f) A notification about the internship is sent to the students whose profile match the skillset and their preferred location
Exit Condition	The User has successfully saved the modifications.
Exception	(d) Some of the required data is missing.

[UC7] Send Message

Name	**Explore**
Actors	Student, Company
Entry Condition	User opens the S&C platform
Event Flow	If a chat already exists: (a) The User opens the **Messages** page (b) The User selects the corresponding chat If a chat didn’t exist yet: (a) The User selects (in the “Explore” section) the profile of the desired recipient (b) The User click on “Send Message” After that, in both cases: (c) The system opens a chat window where the user types a message in the input box and presses the **Send** button.

Exit Condition	The user successfully sends a text message, and it appears in the conversation thread.
Exception	(c) The message fails to send, the system displays an error message: *"Message failed to send. Please try again."* and provides a retry option.

[UC8] Handshake

Name	**Handshake**
Actors	Student, Company
Entry Condition	User opens the S&C platform
Event Flow	(a) The user selected a profile from the Explore page. If the user is a Student, they choose an internship; if the user is a Company, it selects a Student. (b) The system displays the selected profile or project details. (c) The user presses the Send Handshake Request button to initiate contact. (d) The user can add a personalized message along with the request or invitation. If the request is sent by a Company, they must specify the project the request is for. (e) The system sends the request or invitation with the attached message to the selected profile. (f) Upon acceptance, the internship appears in the **My Internships** menu for both parties.
Exit Condition	The request is successfully sent, and both parties can track the request in their respective My Internships menu.
Exception	(d) the Company didn't select the project the request is for, the system displays an error message: *"Please specify the project"* and provides a retry

[UC9] Interviewing

Name	**Interviewing**
Actors	Student, Company

Entry Condition	The internship request has been accepted, and the parties are listed in each other's My Internships menu.
Event Flow	(a) The company initiates the interview process by sending an interview invitation to the student. (b) The system notifies the student about the interview request. (c) The student reviews the invitation and either accepts or declines it. (d) If accepted, the company and the student coordinate the interview details (e.g., date, time, and format). (e) **Optional:** Before the interview, the company may send a questionnaire to the student. ● The system notifies the student about the questionnaire. ● The student completes and submits the questionnaire. ● The system forwards it to the company for review. (f) The system provides tools for conducting the interview (e.g., video call link) or stores interview details if conducted offline. (g) After the interview, the company evaluates the student and updates their status to Accepted or refused.
Exit Condition	The student is either accepted or refused for the internship, and the status is updated in the My Internships menu for both parties.
Exception	(c)(e)(f) Student declines the interview, questionnaire is missing, interview is not attended.

[UC10] Negotiation

Name	**Negotiation**
Actors	Student, Company
Entry Condition	The interview has been conducted, and the student has been accepted for the internship.
Event Flow	(a) The company sends an initial offer or terms to the student. (b) The student reviews the offer and either accepts, refuses, or responds with a counteroffer through a message.

	(c) The system notifies both parties of the offer and any responses (acceptance, refusal, counteroffer). (d) If the student sends a counteroffer, the company can either modify the contract accordingly or refuse. (e) Both parties can continue the negotiation by sending messages through the platform to discuss terms (e.g., salary, work hours, project scope). (f) The negotiation continues until both parties agree on terms, or one party decides to withdraw. (g) Once an agreement is reached, both parties confirm the final terms, and the internship offer is finalized. The system updates the **My Internships** menu.
Exit Condition	Both parties either reach an agreement and finalize the internship offer or one party withdraws, ending the negotiation.
Exception	There is no response from either party within a specified time.

[UC11] Contract Termination

Name	**Contract Termination**
Actors	Student, Company
Entry Condition	The user is logged in and the internship is ongoing.
Event Flow	An internship can end either because it reaches its natural conclusion, or because one of the Users decides to terminate the contract before its scheduled end date. Only if one of the Users wants to terminate the contract: (a) The User selects the corresponding ongoing internship in “My Internships” menu. (b) The User sends a termination request, together with a textual explanation of the decision. (c) The system notifies the other party and the student’s university of the termination request. In both cases:

	(a) The system updates the status of the internship in the "My Internships" menu. (b) The system notifies both parties that the internship is over and that the Satisfaction Survey is available.
Exit Condition	The internship is successfully terminated, and the status is updated in the system.
Exception	(b) The user does not confirm the submission, or doesn’t write anything in the text field.

[UC12] Fill out Satisfaction Survey

Name	**Fill out Satisfaction Survey**
Actors	Student, Company
Entry Condition	Both Users are signed into the platform and the internship is interrupted or completed.
Event Flow	(a) Once the internship reaches its end date or is interrupted, the system prompts both the student and the company to fill out a satisfactory survey. (b) The student completes the survey, rating a set of pre-defined fields and providing textual feedback on their internship experience. (c) The company completes the survey, rating a set of pre-defined fields and providing textual feedback on the student's performance. (d) The system collects both surveys and stores the results. (e) Upon completion of the surveys, the system updates the internship status to "Completed".
Exit Condition	Both the student and company have filled out the satisfaction surveys, and the internship is marked as "Completed."
Exception	

[UC13] Send Complaint

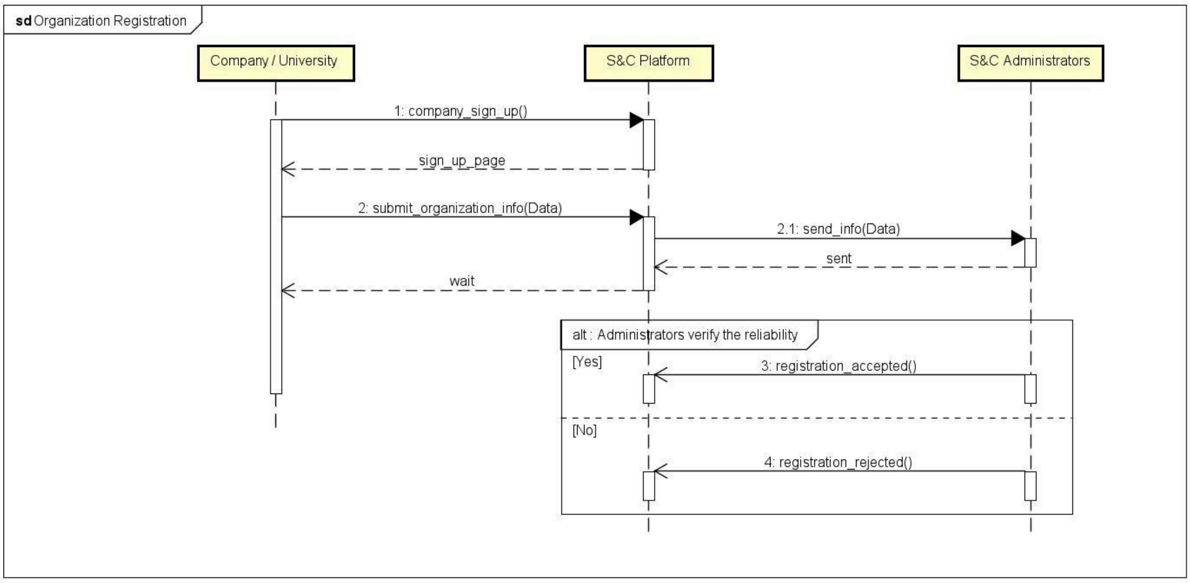
Name	**Send Complaint**
Actors	Student, Company

Entry Condition	The user is logged in the platform, and the internship must be ongoing.
Event Flow	(a) The user who wishes to file a complaint accesses the corresponding internship page in "My Internships" menu (b) The system shows the option to send a complaint to the university regarding that internship (c) The user writes an explanation of the problems encountered in the text box, and presses the "send" button (d) The system acknowledges receipt of the complaint and notifies the university about the complaint submission.
Exit Condition	The complaint is successfully sent to the university
Exception	Invalid User Connection or Incomplete Submission

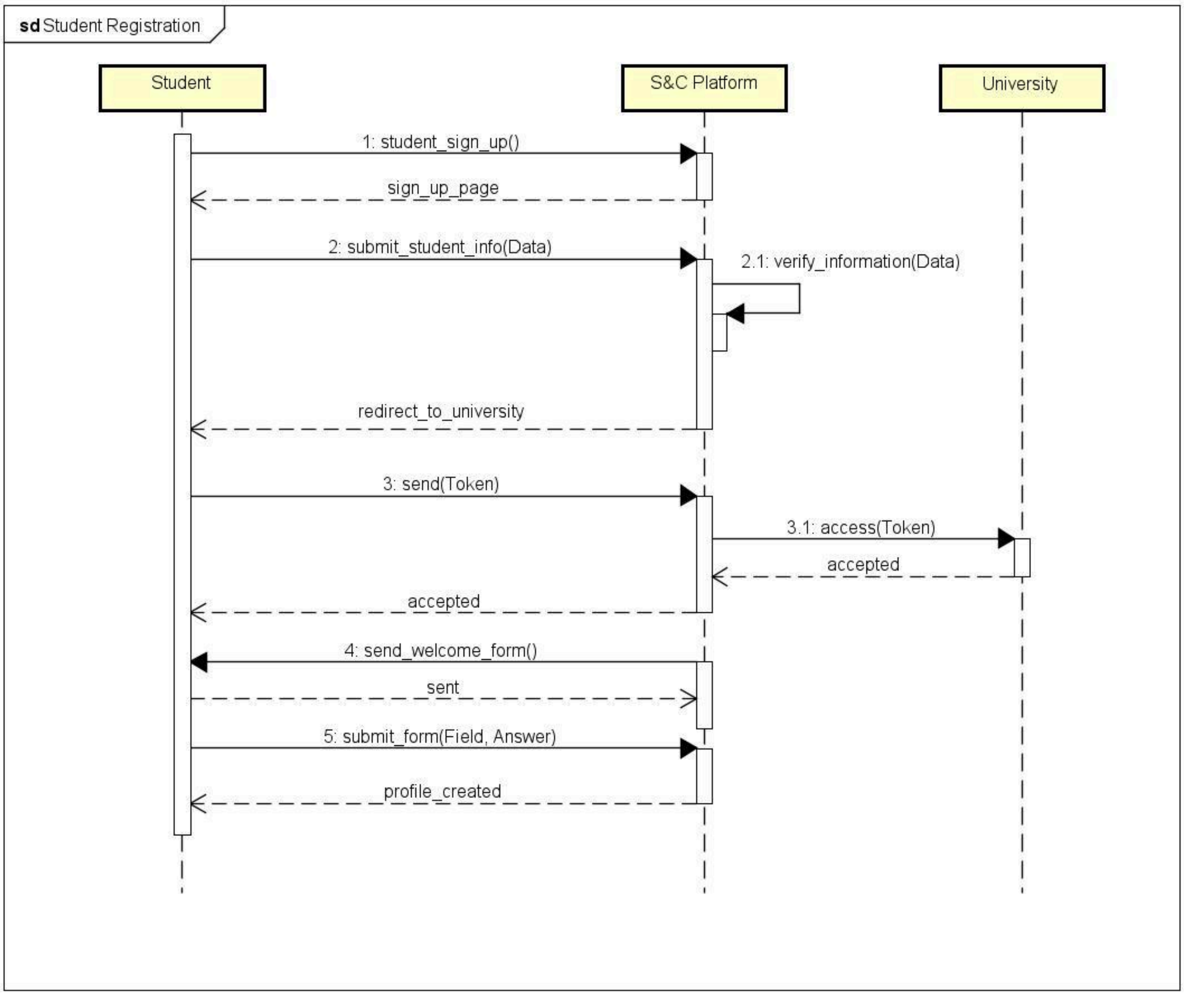
3.2.3 Sequence Diagrams

Please note that the notifications here are modeled at a high level, without considering design and architectural aspects, and omitting synchronization details. They have been introduced just to represent the interaction with users.

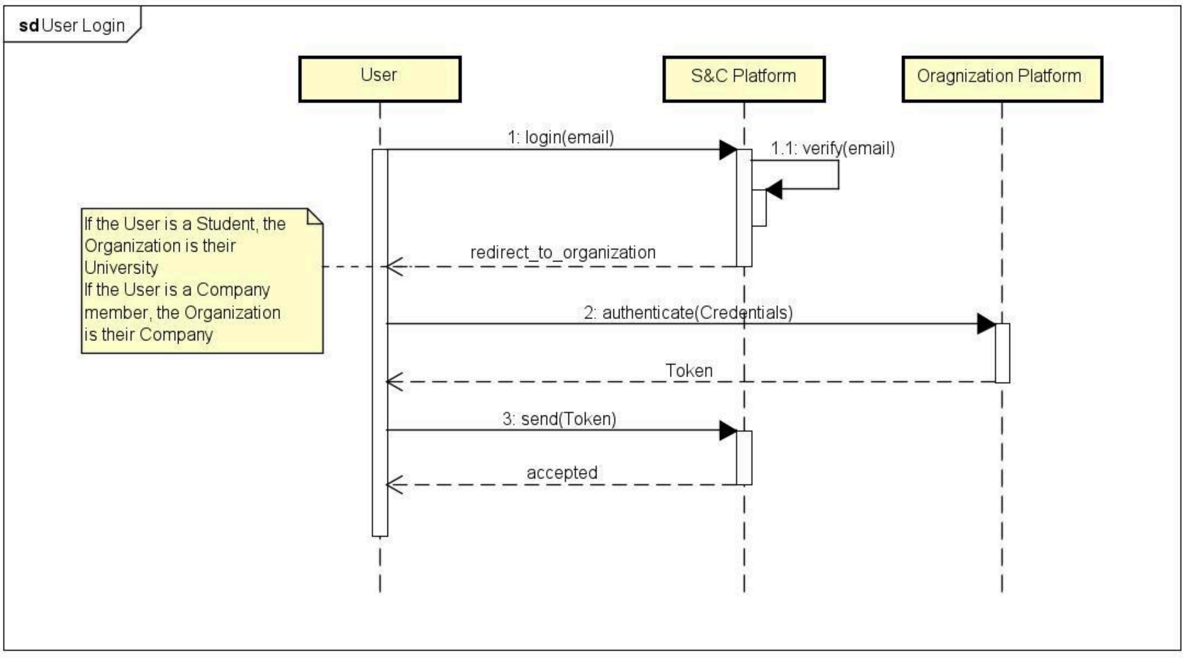
[UC1] Organization Registration



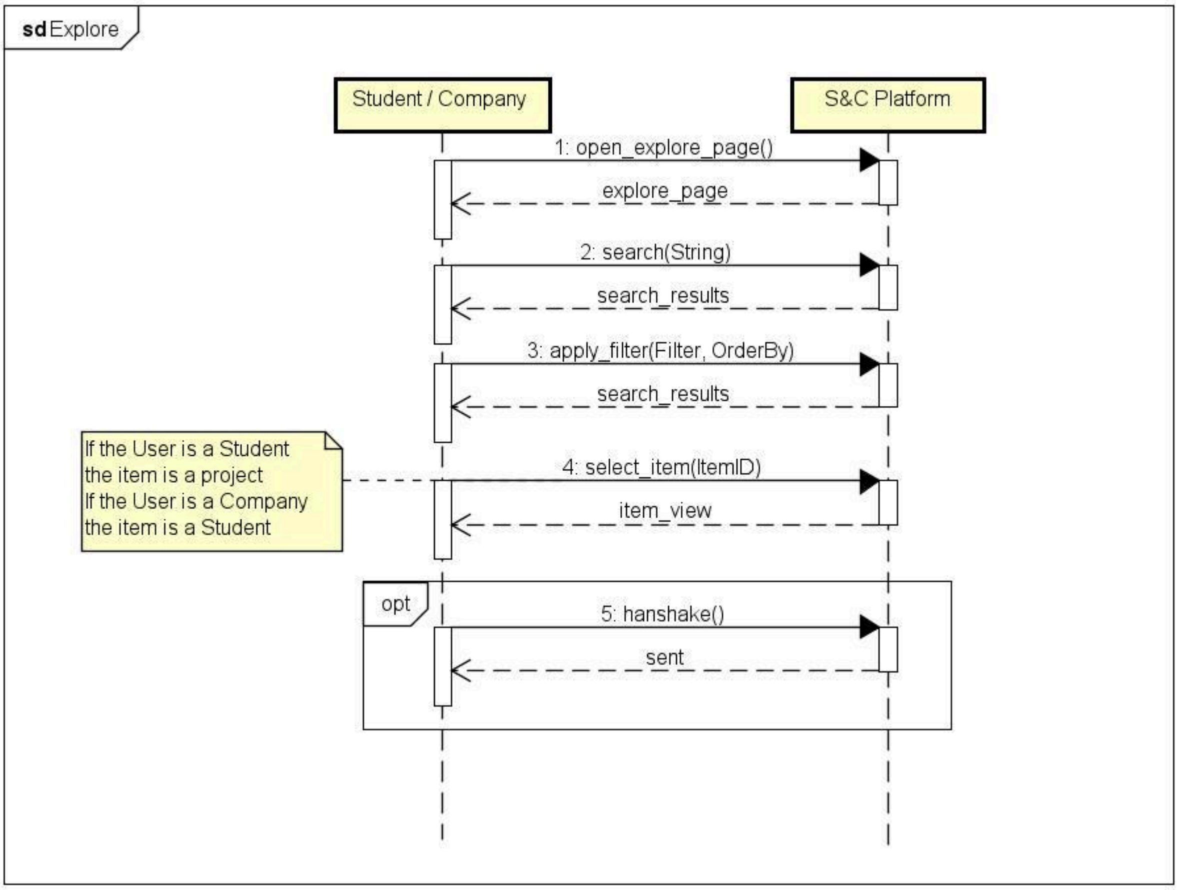
[UC2] Student Registration



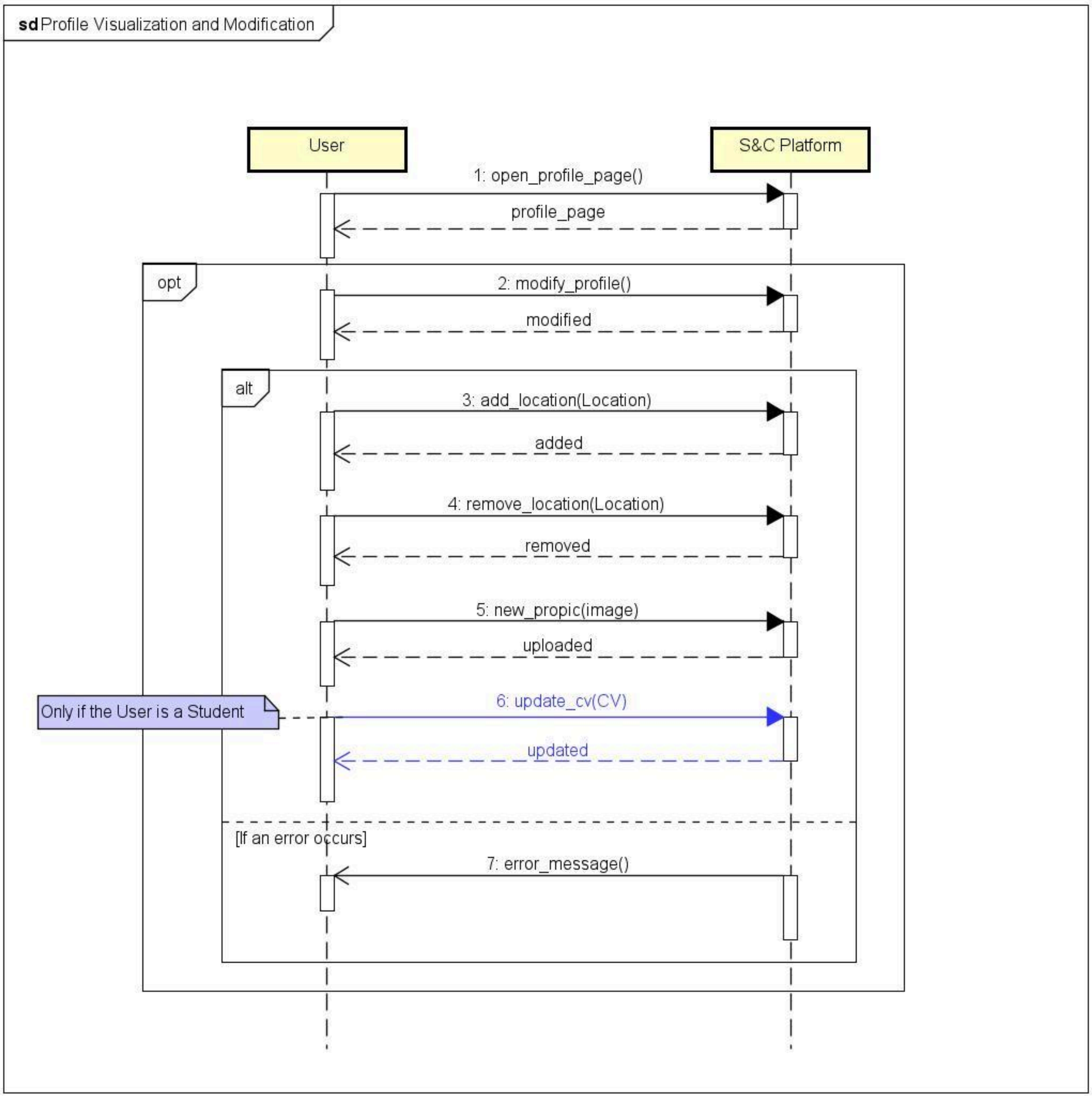
[UC3] User Login



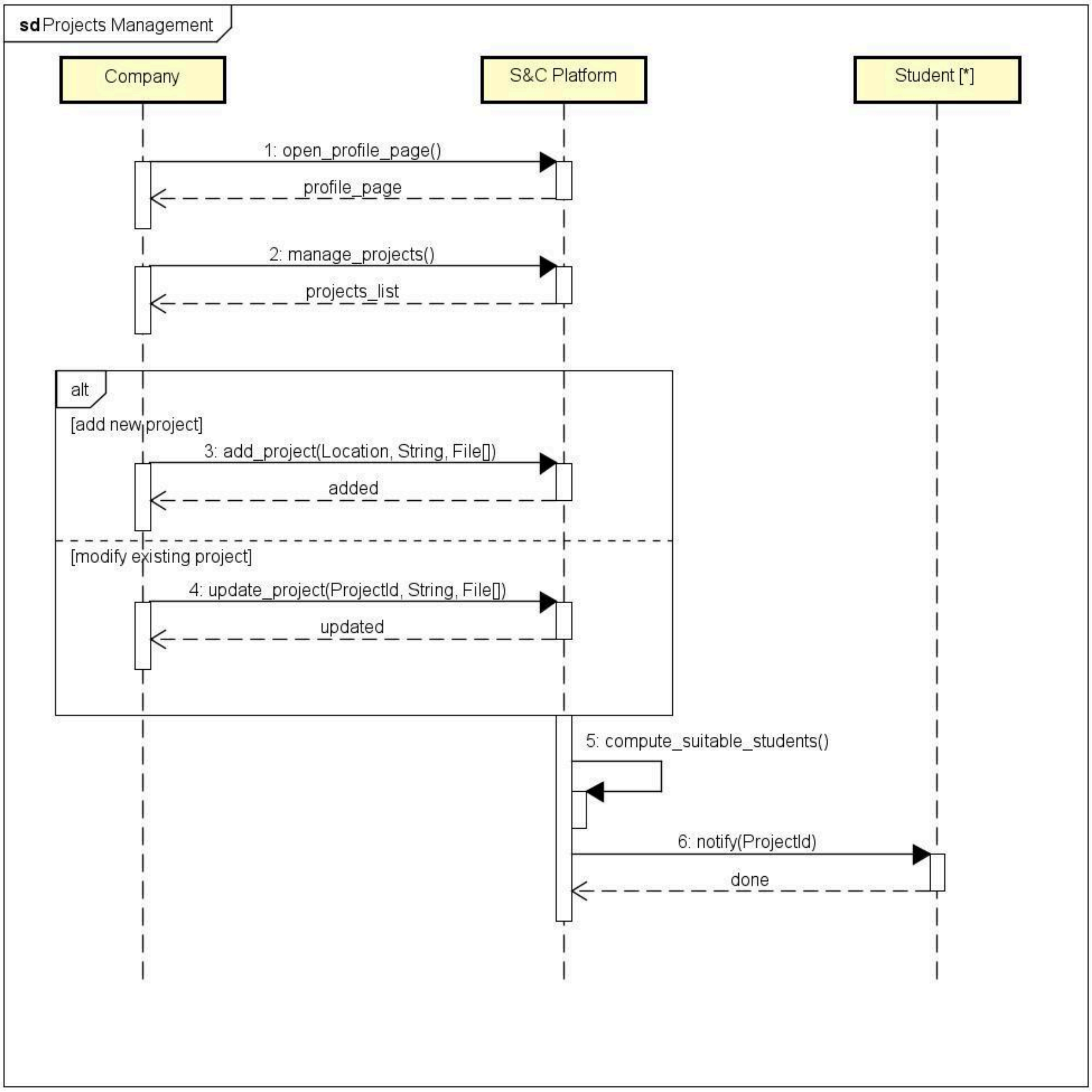
[UC4] Explore


sd Explore
Student / Company
S&C Platform
1: open_explore_page()
explore_page
2: search(String)
search_results
3: apply_filter(Filter, OrderBy)
search_results
If the User is a Student
the item is a project
If the User is a Company
the item is a Student
4: select_item(ItemID)
item_view
opt
5: hanshake()
sent

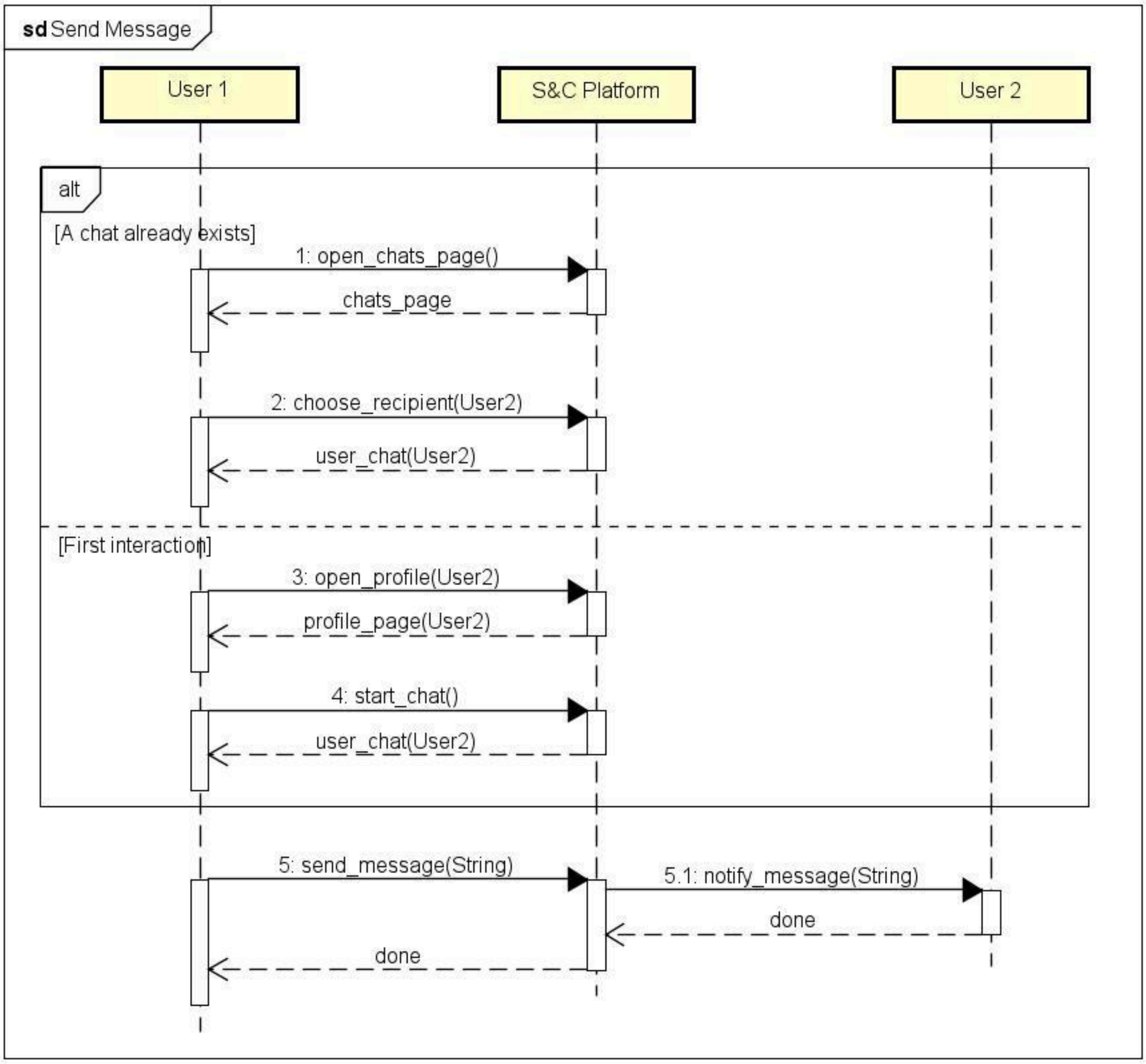
[UC5] Profile Visualization & Modification



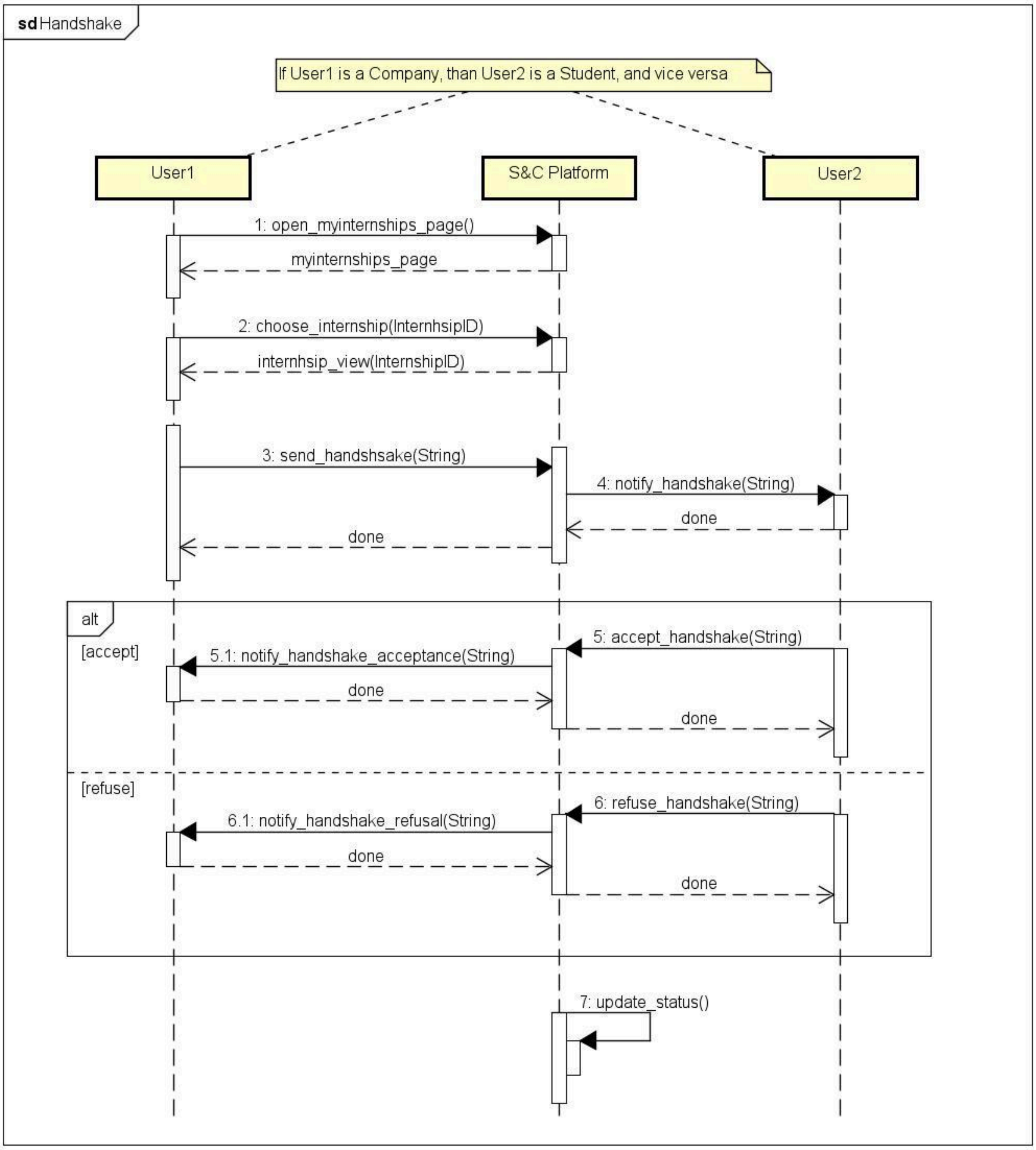
[UC6] Project management



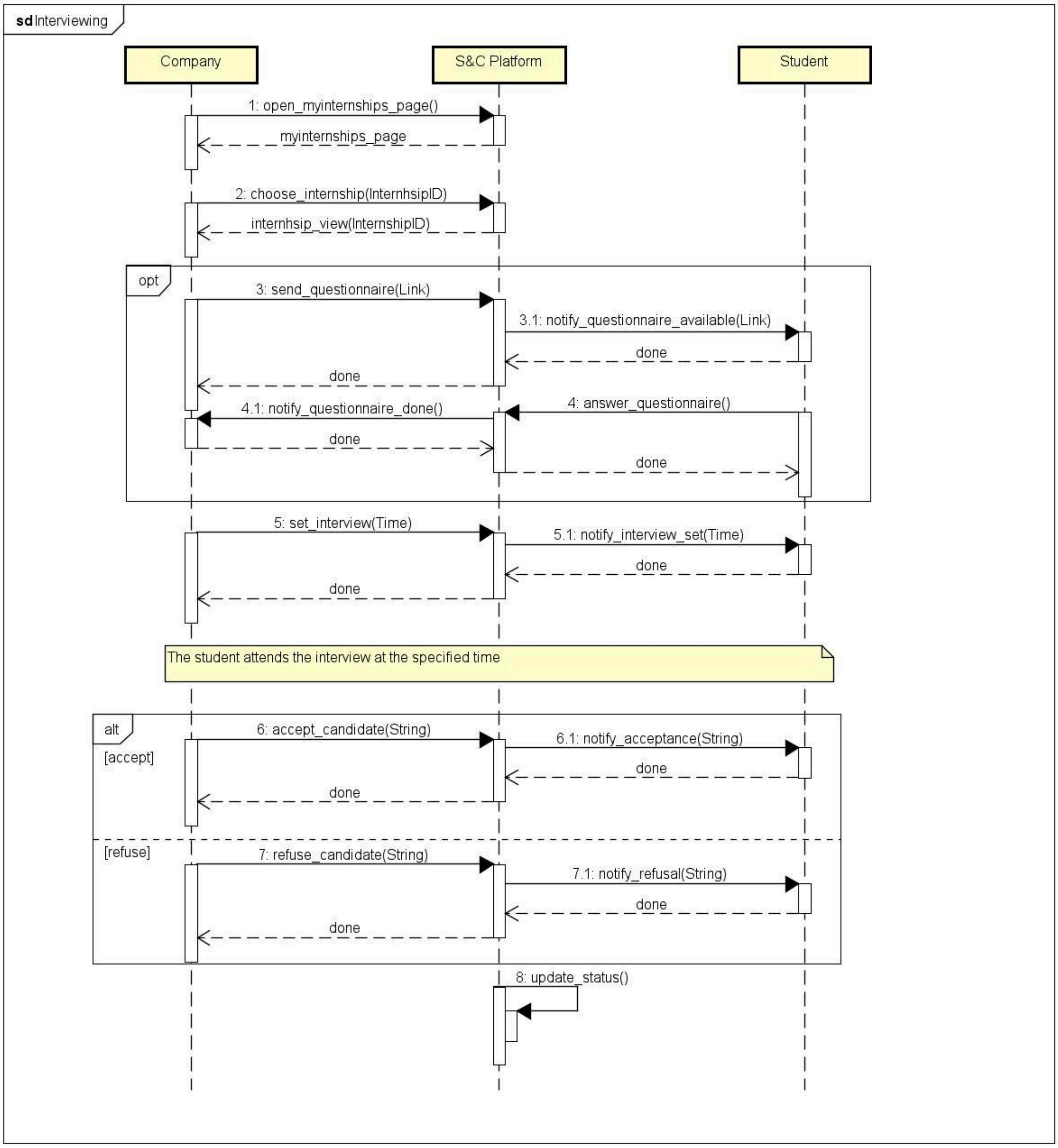
[UC7] Send Message



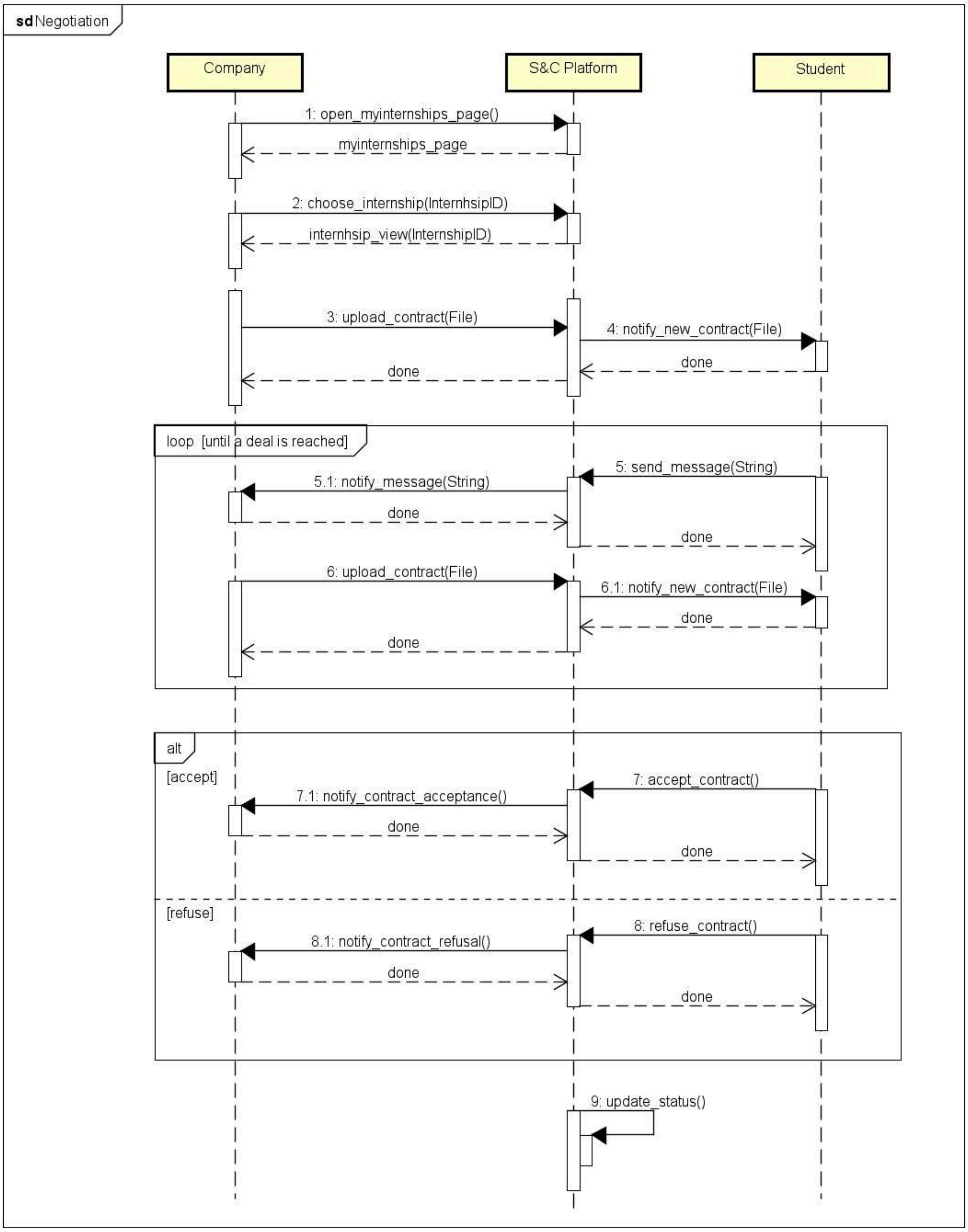
[UC8] Handshake



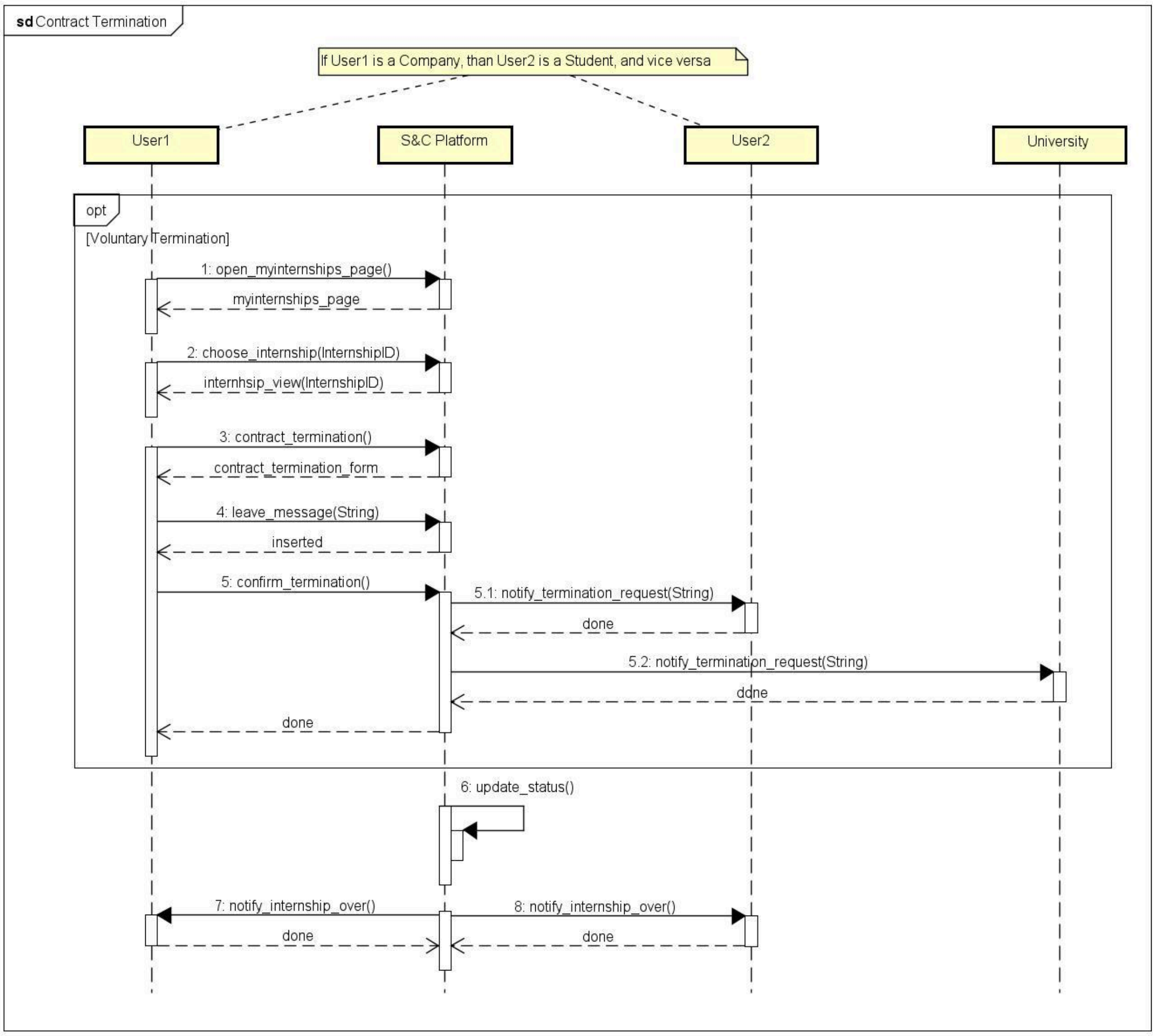
[UC9] Interviewing



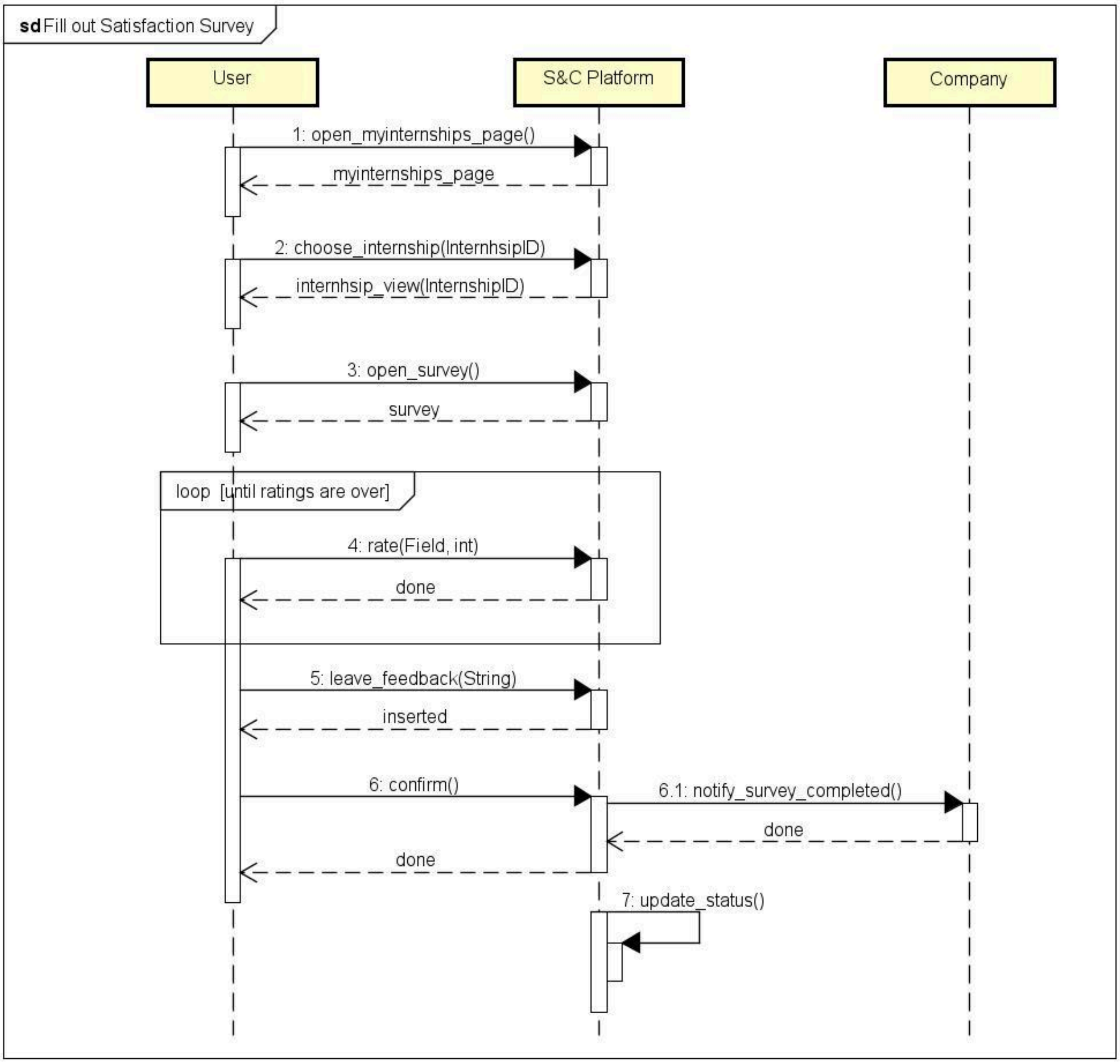
[UC10] Negotiation



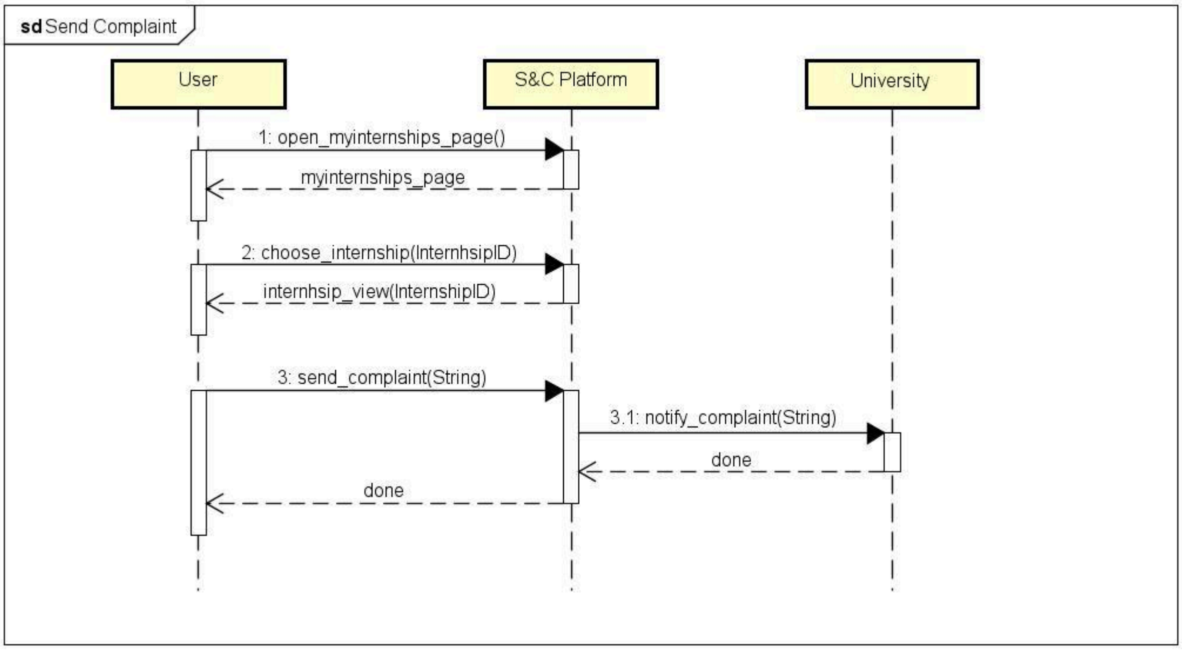
[UC11] Contract Termination



[UC12] Fill out Satisfaction Survey



[UC13] Send Complaint



3.2.4 Requirement Mapping

G1: University students are assisted in finding projects and internship terms offered by companies that best match their skills, experiences, and interests, and in presenting themselves effectively to the companies they wish to engage with.	
R1: The platform allows university students to create an account. R5: The platform allows registered users to log in. R7: The platform allows students to upload and publish their CVs. R8: The platform offers suggestions to students to enhance the appeal of their CV. R9: The platform allows companies to publish available project offers along with all related details and information. R11: The platform allows students to proactively look for internships and provides	D1: Students' personal information is accurate. D3: Students write their CVs, which contain correct information and include the description of their experiences, skills and attitudes. D4: Companies provide detailed information about the projects they offer and related contract terms for students. D5: Students and companies check the recommendations provided by the system and use the platform to accept suitable matches. D10: All users have a reliable internet connection

filters to help them refine their searches. R12: The platform analyses the submissions and computes recommendations that presents to both parties. R13: The system notifies students when a new project that could interest them gets published on the platform by a company. R14: The system notifies companies when a new CV that could interest them gets published on the platform by a student. R15: The platform allows students to visualize companies' profiles. R16: The platform allows companies to visualize students' profiles. R17: The platform allows students to search for job listings based on their skills, interests, and career goals. R24: The platform collects feedback and suggestions from both students and companies and uses this information to feed the analysis mechanisms used for the recommendations.	

G2: Companies publish the projects and internship terms they offer and receive assistance in promoting these opportunities based on their professional fields. They are also supported in connecting with university students who might be suitable candidates for these roles.	
R2 : The platform allows companies to create an account. R4: The platform is able to verify the reliability of its registered users. R5: The platform allows registered users to log in. R7: The platform allows students to upload and publish their CVs. R9: The platform allows companies to publish available project offers along with all related details and information. R10: The platform offers suggestions to companies to refine the presentation of their projects. R12: The platform analyses the submissions and computes recommendations that presents to both parties. R13: The system notifies students when a new project that could interest them gets published on the platform by a company. R14: The platform allows students to visualize companies' profiles. R18 : The platform allows companies to discover potential candidates whose profiles align with job requirements, skills, and qualifications for each project. R24: The platform collects feedback and suggestions from both students and companies and uses this information to feed the analysis mechanisms used for the recommendations.	D2: S&C administrators are able to verify through external checks whether companies and universities requesting to register are reliable. D3: Students write their CVs, which contain correct information and include the description of their experiences, skills and attitudes. D4: Companies provide detailed information about the projects they offer and related contract terms for students. D5: Students and companies check the recommendations provided by the system and use the platform to accept suitable matches. D10: All users have a reliable internet connection.

G3: Companies communicate with students and subject them to selection processes for accessing the projects and terms they offer.	
R7: The platform allows students to upload and publish their CVs. R16: The platform allows companies to visualize students' profiles. R19: The platform allows companies to submit questionnaires to students they are interested in and organize interviews with them. R20: The platform allows companies to offer contracts to students they wish to engage in one of their projects. R21: The platform enables direct communication between students and companies through a messaging system. R23: The platform provides students and companies with tools for tracking and monitoring the matchmaking processes and the internships they are involved in.	D1: Students' personal information is accurate. D3: Students write their CVs, which contain correct information and include the description of their experiences, skills and attitudes. D6: Companies prepare structured questionnaires and interviews to which they subject students. D7: Students and companies are open to communicating and providing feedback on each other. D10: All users have a reliable internet connection.

G4: Students and companies can provide feedback on the matchmaking process and the internship and communicate issues while the projects are ongoing, allowing universities to address them when needed.	
R3: The platform allows universities to create an account. R21: The platform enables direct communication between students and companies through a messaging system. R22: The platform allows companies and students to withdraw from the selection process and internship phase at any time. R26: The platform equips universities with tools to monitor matchmaking processes and the internships in which their enrolled students	D7: Students and companies are open to communicating and providing feedback on each other. D8: Students are enrolled at the university they specify when registering to the platform and are provided with a personal university email. D9: Universities monitor internships and handle complaints that may require their interruption.

participate. R27: The platform provides dedicated spaces where universities can view any complaints from both companies and related students about ongoing internships.	D10: All users have a reliable internet connection.

3.3 Performance Requirements

One of the main performance aspects of the system is its ability to support a large number of registered and connected users, scalability is therefore one of the most critical functional requirements that must be demanded from the system. Given the many expected students who will register on the platform, it must be able to scale easily, minimizing at the same time performance loss. Another key aspect is availability: given the "search engine" nature of the system, it must be easily accessible to its users and provide requested information quickly and directly. Furthermore, mechanisms such as the search, recommendations, and messaging subsystems must be highly responsive and not create bottlenecks. Finally, the system must be capable of supporting the upload of large amounts of documents and data by both students and companies.

3.4 Design Constraints

3.4.1 Standards compliance

Conformity Standards

- GDPR: The system manages users' personal and administrative data, thus it must be compliant with the General Data Protection Regulation according to the European guidelines.
- ISO/IEC 27001: The system must be characterized by a structured and comprehensive information security management, in order to protect sensitive information and be adaptable to new risks.

Development Standards

- ISO/IEC 12207: The system must be compliant to the software lifecycle management standard by applying the best practices in all the main processes that bring to its realization.

3.4.2 Hardware limitations

All users access the system through a web application, therefore hardware limitations are not overly restrictive, even if still present and necessary for the interaction to be smooth and efficient:

- The device's hardware features needed to access the system are strictly dependent on the browser used. According to average browser requirements, the device would need an Intel Pentium 4 cpu or later and at least a 4GB RAM. For being able to correctly visualize the User Interface and utilize its features, the minimum display quality required is HD.
- The interaction between the user and the platform is strictly dependent also on the user's internet connection, which must be stable and have a speed of at least 5 Mbps for loading pages and transferring data quickly.

3.4.3 Any other constraint

The browser used should be up-to-date and compatible with the recent web application standards. Furthermore, it should enable JavaScript and Cookies as advanced functionalities.

3.5 Software System Attribute

3.5.1 Reliability

The system must be capable of preserving a large volume of data retrieved from users and produced by the services that it offers, thus it should **replicate** it across multiple nodes to prevent data loss, and ensure **consistency** across all replicas. The system should employ a robust backup solution and establish a **recovery** procedure to recover lost data. Finally it should be provided with a secure **backdoor** access for administrators for ensuring emergency maintenance, and its usage should be heavily monitored.

3.5.2 Availability

The S&C platform should deliver a highly available service that can be accessed quickly and efficiently by its users. Core functionalities of the system should incorporate **redundancy** and **failover** mechanisms to maintain service continuity. A significant increase in users could cause the system to scale rapidly, adding complexity and potentially reducing service availability. This aspect should be carefully considered during the initial design phase.
Generally, the design should guarantee at least **99,9%** of uptime. It should also implement **circuit breakers** to prevent cascading failures, and administrators should be able to respond quickly and effectively to issues.

3.5.3 Security

The system does not use user passwords, however, it still retains emails and any personal and administrative information provided by users. All stored data must be protected with encryption, and data in transit must also be encrypted to ensure security. The system may be particularly susceptible to Injection and Denial of Service attacks, hence it should incorporate measures to prevent or reduce their impact.

3.5.4 Maintainability

The components should be designed to be easily scalable, maintainable, reusable and testable. To achieve this, the system should implement modular design principles and ideally adopt a microservice architecture. The code should be readable, maintainable and fully documented. Documentation should include detailed and well analyzed API definitions, a comprehensive description of the system architecture and should define the testing strategies employed.

3.5.5 Portability

As previously explained, the system should be compatible with the majority of web browsers, utilizing both their primary and supplementary features. Moreover, the platform should be optimized also for use on smartphones and tablets, allowing for switching web views and ensuring flexibility across the various devices used to access it.

4. Formal Analysis Using Alloy

In this Alloy analysis, we chose to focus on internships. Specifically, we want to describe the evolution of selection processes over time, the fulfillment of minimum skill requirements for each project, and the capacity constraints of each project. To achieve this, we modeled the Internship entity as mutable, with its current step progressing over time.

We point out that “Step 0” is not part of this model, as previously explained, it is a "symbolic" state that does not represent anything other than the fact that no contact has yet occurred between a given student and a specific project offered by a company.

```
sig Student {
    var internships: set Internship,
    attends: one University,
    skills: set Skill
}

// Every university must have at least one student attending it
sig University {} {
    some s: Student | s.attends = this
}
```

```
// Only companies organizing at least one project are considered
sig Company {} {
	some p: Project| p.organizedBy = this
}

// A project has a maximum capacity for internships and may have a minimum set of skills required for students to be eligible
sig Project {
    organizedBy: one Company,
    capacity: one Int,
    minimumSkills: set Skill
} {
    capacity > 0
}

// An internship is linked to a single project and a single student
// It goes through several steps of a selection process before becoming definitive
// Internships are mutable: they can start or end at any time, and their current step can evolve over time
var sig Internship {
    var ofProject: one Project,
    var currentStep: one Step
} {
    one s: Student | this in s.internships
}

sig Skill {}

enum Step { Step1, Step2, Step3, Step4 }

// At any moment, the number of internships related to a project cannot exceed the project's capacity
fact CapacityConstraint {
    always (all p: Project | #(ofProject.p & currentStep.Step4) <= p.capacity)
}

// At all times, students participating in the selection process must meet the project's minimum skill requirements
fact MinimumRequiredSkills {
	always (all i: Internship | all s: internships.i | s.skills in i.ofProject.minimumSkills)
}

// This function returns the step that follows the given step as input
fun nextState[i: Internship]: one Step {
    (i.currentStep = Step1) => Step2 else
    (i.currentStep = Step2) => Step3 else
```

```
    (i.currentStep = Step3) => Step4 else
    (i.currentStep = Step4) => Step4 else none
}

// The project associated with an internship never changes
fact SameProject {
        always (all i: Internship | (i in Internship') implies i.ofProject' =
i.ofProject)
}

// Describes the progression of internships: they either move to the next step,
stay at the current step, or are terminated
fact StateProgression {
    always (
        all i: Internship |
        i.currentStep' = i.currentStep or
        i.currentStep' = nextState[i] or
        i not in Internship'
    )
}

// All internships must start at Step1
fact NewInternship {
    always (
        all i: Internship | once (i.currentStep = Step1)
    )
}

// A student cannot participate in more than one internship organized by the same
company at the same time
fact OneInternshipPerCompany {
    always (
        all s: Student |
                  no disj i1, i2: s.internships | i1.ofProject.organizedBy =
i2.ofProject.organizedBy
    )
}

// Every internship eventually ends, and once interrupted, it cannot be resumed
fact EndInternship {
    always (all i: Internship | eventually (i not in Internship))
     always (all i: Internship | i not in Internship' implies after always i not in
Internship)
}
```

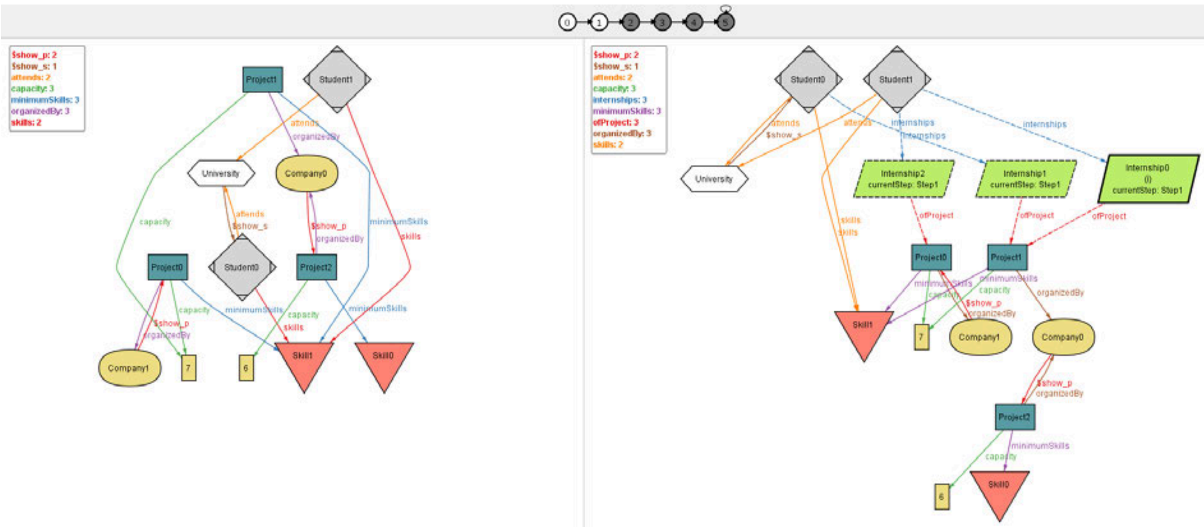
Now, let's look at some specific examples to have a better understanding of how the model represents the domain of our system.

ex. 1

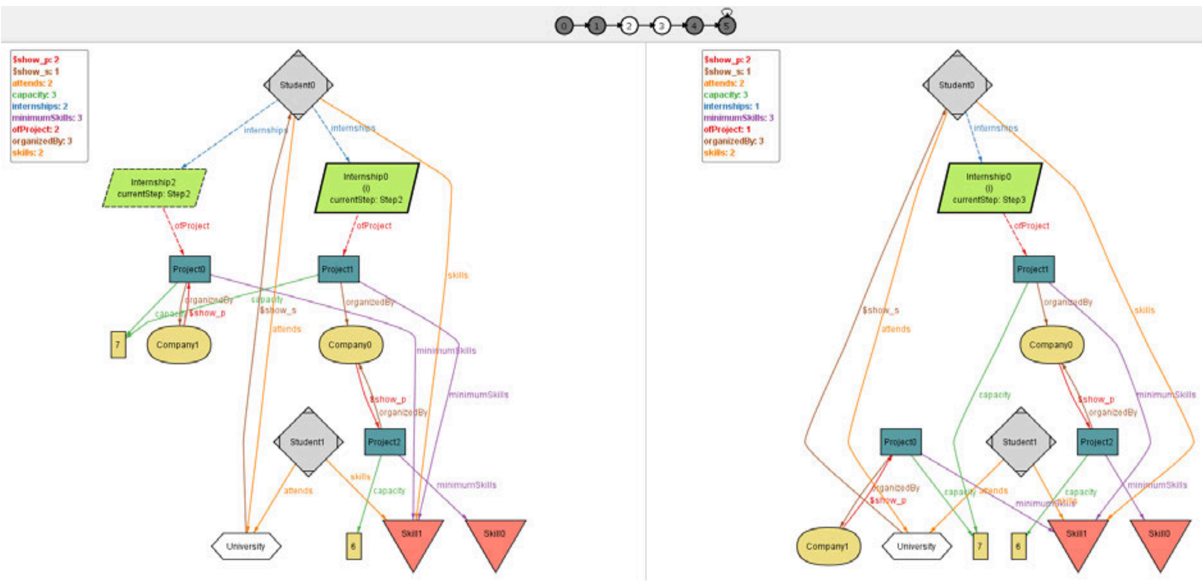
```
pred show[] {
        #Internship = 0
    eventually (#Internship = 2)
    eventually (some i: Internship | i.currentStep = Step4)
}

run show for 3 but exactly 2 Student
```

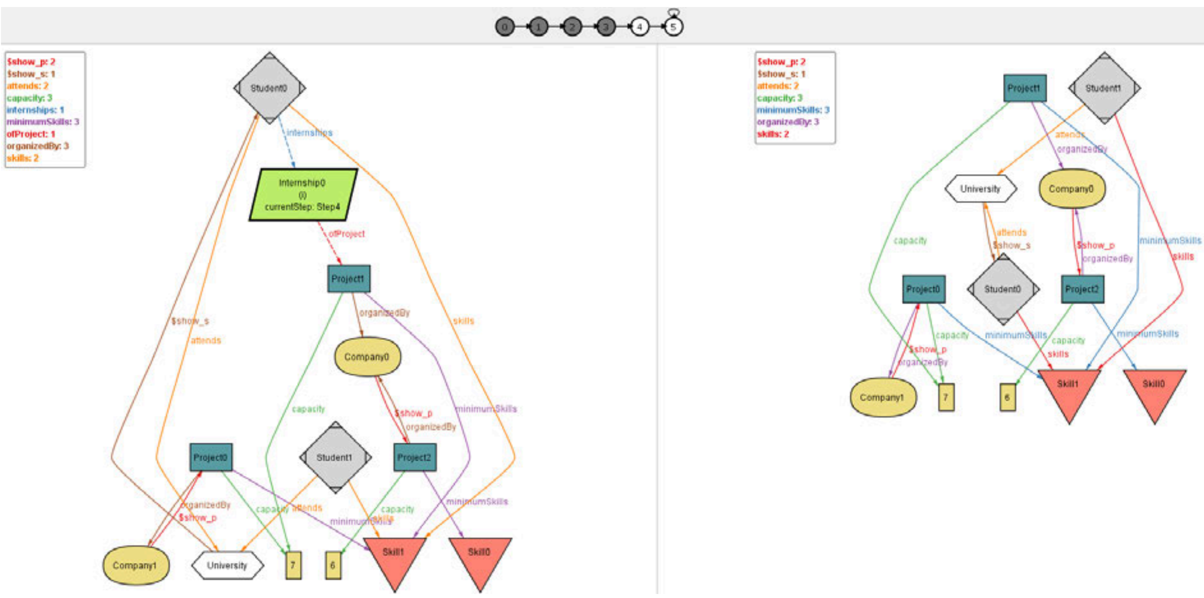
In this example we can see a typical scenario. There are two selection processes, and while one of them arrives at Step4 (meaning that the selection has been successful), another one stops at Step3.



At the beginning, no internships existed. Then, Student0 and Student1 initiate one and two selection processes, respectively. Both students meet the minimum skill requirements.



Internship1 is interrupted at Step1, while the other two progress to Step2. In the following instant, only Internship0 remains active and advances to Step3.



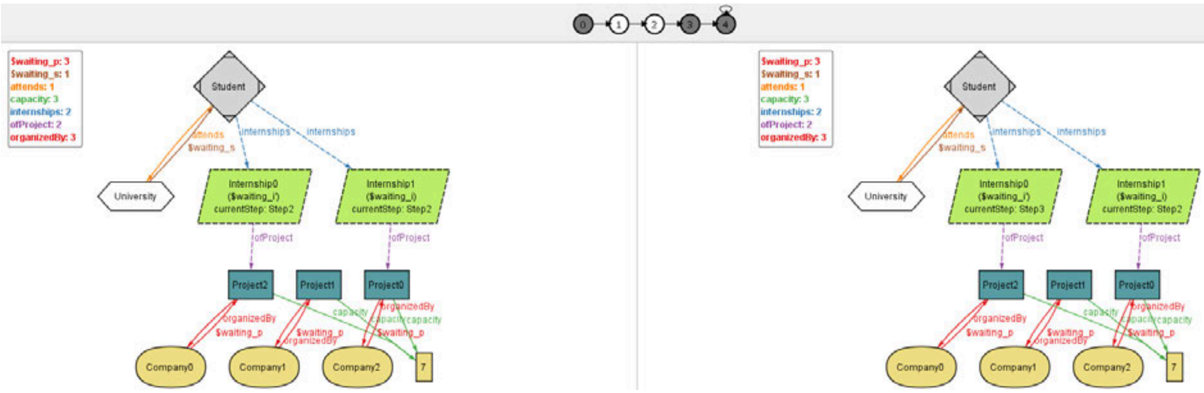
The selection process concludes when Internship0 reaches Step 4. In the subsequent instant, its termination is represented.

ex. 2

```
pred waiting[] {
	//skills are hidden to improve readability
	#Skill = 0
	eventually(some i: Internship| i.currentStep' = i.currentStep)
	eventually(some i: Internship | i.currentStep = Step4)
}

run waiting for 3 but exactly 2 Student
```

A Step may require more time to complete, which is modeled by allowing internships to remain in the same state for multiple instants.



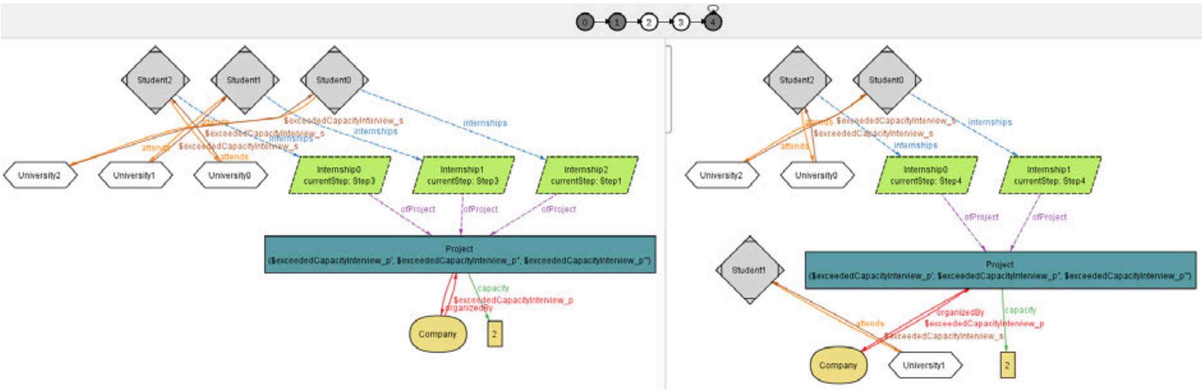
In this example, Internship1 remains at Step2 for two instants, while Internship0 progresses.

ex. 3

```
pred exceededCapacityInterview[] {
	//skills are hidden to improve readability
	#Skill = 0
	some p: Project| p.capacity = 2
	eventually(
	some p: Project| #(ofProject.p) > p.capacity
	)
	eventually(
	some p: Project| #(ofProject.p) = p.capacity and
	all i: ofProject.p| i.currentStep = Step4
	)
}

run exceededCapacityInterview for 4 but exactly 1 Project
```

As shown in the model, the capacity constraint only considers ongoing internships (Step4). This means that a company could potentially interview a large number of students but must eventually select the best candidates to hire without exceeding capacity. The following example illustrates this scenario.



ex. 4

```
assert noMissingStep {
    always not(
        some i: Internship |
        i.currentStep = Step4 and (
            before historically not(i.currentStep = Step3) or
            before historically not(i.currentStep = Step2) or
            before historically not(i.currentStep = Step1)
        )
    )
    always not(
        some i: Internship |
        i.currentStep = Step3 and (
            before historically not(i.currentStep = Step2) or
            before historically not(i.currentStep = Step1)
        )
    )
    always not(
        some i: Internship |
        i.currentStep = Step2 and (
            before historically not(i.currentStep = Step1)
        )
    )
}

check noMissingStep for 8
```

Finally, through an assertion, we ensure that every internship reaching a certain step has completed all preceding steps in the correct sequence.

```
Executing "Check missingStep for 8"
   Solver=sat4j Steps=1..10 Bitwidth=4 MaxSeq=7 SkolemDepth=1 Symmetry=20 Mode=batch
   1..10 steps. 525345 vars. 15350 primary vars. 1212404 clauses. 11317ms.
   No counterexample found. Assertion may be valid. 3449ms.
```

5. Effort Spent

The following table displays the number of hours each group member spent on the various sections of the document. Please note that the division is only approximate, and each section still required the collaboration of all members.

	Chapter 1	Chapter 2	Chapter 3	Chapter 4
	1	12	25	2
	3	14	17	9
	6	11	23	1

6. References